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Representation Before Training: A Practical Benchmark for Generative Medical Event Model Tokenization

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Abstract

Generative medical event models use tokenized sequences of patient timelines as input, but practical guidance on the many decisions around tokenization is limited. We benchmark quantization granularity, reference-range anchoring, code–value fusion, numeric and temporal encodings, and native versus harmonized event representations from an expert-mapped common data model. Using both Llama and Qwen architectures, 156 models were trained on full hospitalizations from three initialization seeds, with each configuration following a shared training recipe for up to five epochs. We evaluated learned representations from the first 24 hours of hospitalization with linear probes to predict binary and continuous outcomes during hours 24-48. Fused tokens pairing codes with value deciles increased performance across all eight outcome families relative to the equivalent unfused tokenized input with area under the receiver operating characteristic curve (AUROC) gains of +0.002 to +0.033 and Spearman correlation gains of +0.025 to +0.114. Neither anchoring value bins to reference ranges nor increasing the granularity of quantization consistently improved performance, while xVal variants underperformed both discrete and soft encodings. Alternatives to explicit time tokens such as event order and admission-relative rotary position embeddings (RoPE) yielded higher family-mean point estimates across all eight families while reducing the input length. When evaluating native input against input mapped to the Common Longitudinal Intensive Care Unit Data Format (CLIF), the CLIF full-hospitalization training sequences contained 28.6% as many tokens as the native sequences, while improving performance in six of eight outcome families. These findings show tokenization and event encoding are consequential design choices when learning patient representations for downstream classification and regression.

Keywords: Electronic health records, Generative medical event models, Input representation, Quantization, Transformer architectures

Data and Code Availability We use the MIMIC-IV dataset (Johnson et al., 2023), version 3.1, distributed through PhysioNet (Johnson et al., 2024), subject to data use requirements. For MIMIC-to-CLIF schema remapping, we use the CLIF 2.1.0 conversion of MIMIC-IV (Liao et al., 2026). All source code to reproduce the experiments in this paper is available at: `bbj-lab/input-representation-benchmark`

Institutional Review Board (IRB) We used the MIMIC-IV database in accordance with PhysioNet terms so no additional IRB approval was required.

1. Introduction

Electronic health records (EHRs) record patient trajectories as a sequence of clinical events, measurements, and interventions. Generative medical event models are a class of foundation models that learn from these sequences through next-token prediction, yielding representations that can be used for downstream prediction tasks. Before training begins, there are a great number of considerations to be made regarding how clinical data are turned into tokenized sequences. In a standard workflow, events captured by the elec-

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tronic health record are converted into a sequence of tokens. Every recorded event (e.g., a lab result, the taking of vitals, the administration of medication) is mapped to a token or series of tokens and inserted into a patient's sequence in chronological order. A transformer model is then trained to predict the next token in a sequence given all previous tokens. The tokenization of numeric values, representation of timing information, and the granularity of token vocabulary impact sequence length and even the framing of the prediction problem itself.

Representation choices are consequential but incompletely characterized. Structured records combine categorical events with numeric measurements, so the input representation determines how magnitude, elapsed time, and code identity enter the model. Numeric values are commonly discretized into population quantiles, split by clinical reference ranges, or encoded as continuous values. Discretized values can appear as separate tokens or can be fused with the accompanying code token which specifies what was being measured. Discretization does not always align with clinical utility; for example, potassium values of 5.5 and 7.0 mEq/L both exceed the usual upper reference limit but represent different severities of hyperkalemia and electrophysiologic risk (Weiss et al., 2017). Decile-based quantization is likely to group the above range into just one bin, and will also often represent abnormal values with the same token used for some values that are "normal" according to the reference range. Information around event timing can be provided by explicitly inserting timing tokens or through positional encoding. To enable multi-institutional research, many common data models have emerged to provide standardized clinical schemas in an attempt to harmonize heterogeneous data sources. In this work, we leverage the Common Longitudinal Intensive Care Unit Data Format (CLIF) 2.1.0 (Rojas et al., 2025). All of these choices change the way clinical information is expressed when training the transformer model. Information discarded during tokenization cannot be recovered by the transformer that follows, yet most benchmarks hold tokenization fixed while varying architecture, objective, or scale.

The primary objective of this work is to identify useful starting configurations and to characterize the tradeoffs of decisions made during tokenization under a shared training recipe. Concurrent work isolates three binary representation choices across pediatric tasks (Guo et al., 2026); here we add benchmarks of quantization granularity, reference-range anchoring, continuous numeric encodings, regression outcomes, and schema remapping to a common data model for adult MIMIC-IV admissions. We examine this in three primary experiments:

1. **Experiment 1 Quantization**: How should numeric values be quantized, and should category and value share a token? **Finding**: Fused category–value decile tokenization increased mean performance across all eight outcome families relative to unfused deciles, with particularly large gains for numerical forecasting. Finer quantization and reference-range anchoring did not yield consistent improvements, supporting fused deciles as a practical starting configuration.
2. **Experiment 2 Value and temporal encoding**: How should numeric magnitude and elapsed time be encoded? **Finding**: Both tested code-normalized xVal variants underperformed discrete and soft encodings under the shared training recipe. Within discrete encoding, alternatives to explicit time tokens remained competitive while shortening sequences. These findings show that additional encoding complexity did not consistently improve frozen-probe performance in this setting.
3. **Experiment 3 Native MIMIC versus CLIF schema**: How does remapping native MIMIC event streams to an expert-defined domain-specific common data model change downstream prediction performance? **Finding**: The CLIF full-hospitalization training sequences contained 28.6% as many tokens as the native sequences. Mean performance increased in six of eight outcome families, while some individual outcomes declined. This comparison illustrates the robust potential of compact standardized representations and the importance of evaluating their task-specific tradeoffs.

These experiments were performed across 156 training runs spanning both Llama-style and Qwen-style architectures to train decoders across three seeds and a shared optimizer configuration for maximum of five-epochs per decoder with early stopping. Foundation models are trained on full hospitalizations. We evaluate frozen representations of the first 24 hours of hospitalization using linear probes to predict outcomes during hours 24–48. This protocol measures how useful the transformer's learned representation is for predicting a diverse set of critical care outcomes.

2. Related Work

2.1. Generative Medical Event Models and Their Benchmarks

Clinical sequence models have diversified their architectures and objectives while often inheriting a fixed event serialization. Early recurrent systems established prediction from coded histories and timing: Doctor AI jointly predicts next-visit codes and time (Choi et al., 2016a), RETAIN uses reverse-time attention to score hospital outcomes (Choi et al., 2016b), and Dipole applies bidirectional attention to diagnosis prediction (Ma et al., 2017). Self-supervised transformers subsequently showed that pretraining on electronic health record (EHR) sequences transfers to downstream tasks (Li et al., 2020; Rasmy et al., 2021). Foresight (Kraljevic et al., 2024), EventStream GPT (McDermott et al., 2024), and ETHOS (Renc et al., 2024) all adopt causal architectures. Later models vary the architecture or loss: ETHOS-ARES trains on Medical Event Data Standard (MEDS) sequences (MEDS Community, 2024; Renc et al., 2025), EHRMamba replaces attention with a state-space backbone (Fallahpour et al., 2024), MOTOR optimizes a time-to-event objective (Steinberg et al., 2024), and large language models encode serialized event text (Hegselmann et al., 2026)—while Curiosity (Waxler et al., 2025) and Burkhart et al. (2025) retain ten-bin laboratory quantization.

Then, many evaluations likewise embed representational assumptions in their inputs. EHRSHOT evaluates frozen representations on 15 tasks and distributes CLMBR-T-base with laboratory deciles (Wornow et al., 2023), whereas the original CLMBR uses daily code sequences without quantitative laboratory or vital-sign values (Steinberg et al., 2021). HiRID-ICU-Benchmark defines tasks over high-resolution physiology (Yèche et al., 2021); newer suites broaden data coverage through multimodal MIMIC-IV stays (Yu et al., 2025), 44 datasets and 15 modalities (Dai et al., 2025), or prompting over structured EHRs (Yang et al., 2026). Guo et al. (2026) directly compare fused and unfused tokens, interval tokens and age-derived rotary position embeddings (RoPE), and collapsed and expanded workflow events across 74 pediatric tasks, finding a consistent fused-token advantage. Our benchmark extends that controlled comparison to quantization granularity, reference-range anchoring, continuous numeric encodings, regression outcomes, and schema remapping on adult hospital and intensive care unit (ICU) admissions.

2.2. Encoding Magnitude and Time

To encode data as input to a transformer model, EHR data are commonly converted to a chronological sequence of events (MEDS Community, 2024) where each event has a subject id, a timestamp, a code or category, and where appropriate a numeric value (e.g., a lab result). Encoding a scalar inside a discrete-token transformer remains an open problem in natural language processing (NLP) (Thawani et al., 2021) and is especially consequential in EHRs, where clinically distinct values may share a token. In equal-frequency quantization, all numerical training values within each category (e.g., standardized medication type, type of lab, type of vital) are used to determine quantile bins that are then applied to all values associated with a given category. Equal-frequency quantization approximately balances values within each code, subject to tied or rounded measurements, but its boundaries need not align with clinical thresholds (Waxler et al., 2025). xVal injects magnitude by scaling a shared `[NUM]` embedding and adding an auxiliary numeric head (Golkar et al., 2023). ConSE (Convex Combination of Semantic Embeddings) forms convex combinations of category embeddings for zero-shot classification (Norouzi et al., 2014); we adopt its local two-point interpolation between adjacent quantile embeddings. Continuous autoregressive language models instead predict the next continuous vector over compressed token chunks (Shao et al., 2025), while specialized event-time architectures process continuous clinical values without a shared quantile table (Labach et al., 2023; Ma et al., 2024).

EHR transformers encode time through inserted interval tokens and age embeddings (Pang et al., 2021), learned continuous features (Kazemi et al., 2019), or rotary embeddings keyed to days since birth within a time-to-event objective (Steinberg et al., 2024; Su et al., 2024). In a controlled MIMIC-IV ablation across four prediction tasks, removing the explicit time term or replacing it with Time2Vec or LeTE produced no consistent statistically significant improvement (Attrach et al., 2025). Experiment 2 retains the same decoder configurations and optimization settings while crossing four numeric encodings—discrete bins, soft interpolation, code-normalized xVal, and affine xVal—with inserted time tokens, event order, and admission-relative rotary position embeddings at 30-second resolution.

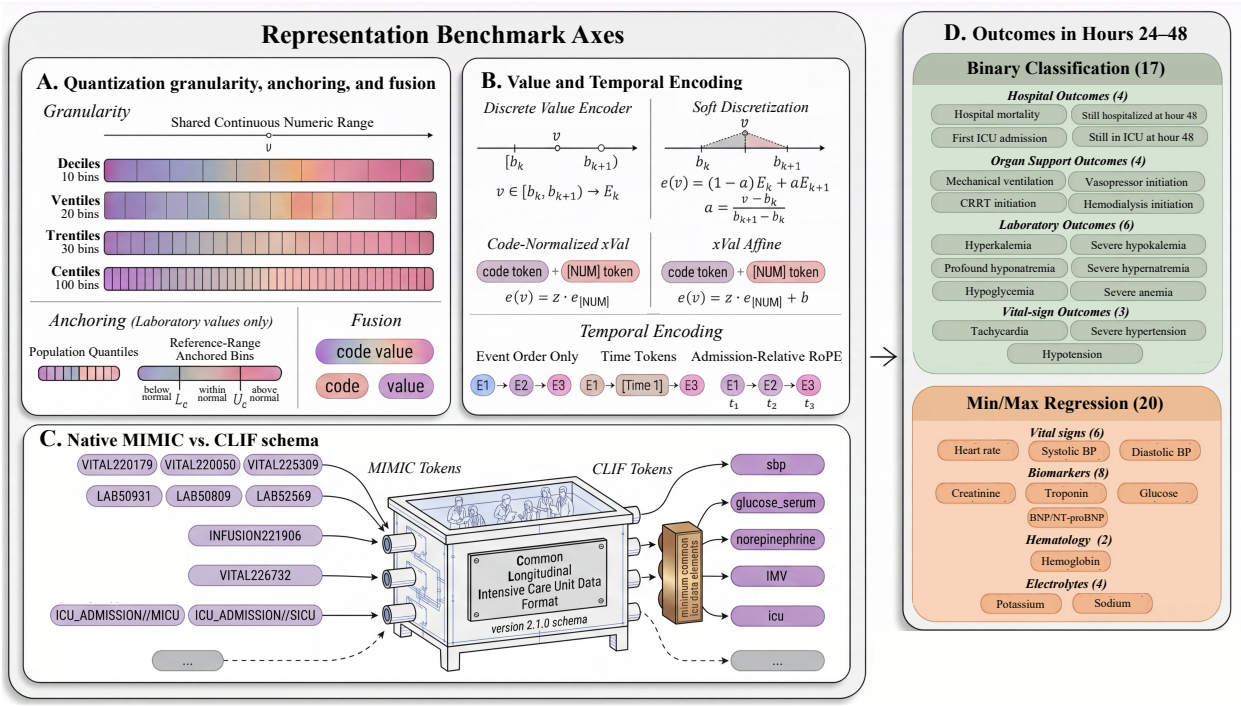

Figure 1. Panels A–C match Experiments 1–3 in order. Panel D is the shared 24–48h evaluation.

2.3. Schema Standards

Schema harmonization changes both which events remain in a sequence and how source codes are grouped across different sites. The Observational Medical Outcomes Partnership (OMOP) common data model supports ontology alignment across health systems (Callahan et al., 2023), while model-facing methods represent code semantics through text descriptions (Hur et al., 2022), subword decompositions of the International Classification of Diseases, Tenth Revision (ICD-10) and Systematized Nomenclature of Medicine Clinical Terms codes (SNOMED CT) (Dwivedi et al., 2024), or multimodal tokens enriched with graph context (Su et al., 2025). Those methods modify individual code representations. Experiment 3 instead compares native MIMIC MEDS streams with end-to-end schema remapping to CLIF 2.1.0 (Rojas et al., 2025; Liao et al., 2026); event selection, code collapse, and token-frequency distributions therefore change together (Section 4.3).

3. Methods

Data and evaluation window. Each hospital admission (`hadm_id`) is represented as a chronological MEDS event sequence(MEDS Community, 2024). MEDS records contain `subject_id`, `time`, `code`, and `numeric_value` field where appropriate. Foundation models are trained on full-hospitalization event sequences. Frozen representations of the first 24 hours are extracted for linear probes that predict outcomes in $(24, 48]$ among admissions eligible at hour 24. Data are partitioned by patient to avoid multiple admissions being split between training/test (Appendix D).

Experiments 1–2 evaluate 17 binary outcomes spanning hospital status, organ support, and physiologic abnormalities, including death, invasive mechanical ventilation initiation, and hyperkalemia. Twenty continuous outcomes comprise the minimum and maximum values of selected vital signs and laboratory measurements during follow-up, including heart rate, potassium, and creatinine (Table 4). Death during follow-up and hospitalization status at hour 48 use all eligible admissions. Physiologic-threshold and intervention tasks exclude all admissions meeting the endpoint during the input window. Intervention tasks are labeled negative when the intervention is not recorded within the prediction window. Physiologic-threshold and regression tasks require a qualifying follow-up

measurement. Hospital and ICU occupancy are assessed at hour 48.

Experiment 3 evaluates hospitalizations involving ICU care at any time, using a separate patient-level split (Appendix D). Native MIMIC and CLIF inputs use identical admission identifiers and labels derived from native MIMIC. Experiment 3 omits the ICU admission task because every admission in that cohort has an ICU stay, and retains the "still in ICU at hour 48" task, yielding 16 binary and 20 regression tasks (Table 4). The ICU-eligible admissions receive a separate patient-level split. Figure 1 summarizes the experiments.

3.1. Representation Choices

Quantization and Fusion. We partition each numeric code's values into 10, 20, 30, or 100 equal-frequency bins. Unfused representations use a separate code token followed by the numeric bin token. Fused tokens use one token per code-numeric bin pair, increasing embedding parameters and vocabulary size. For laboratory measurements, reference-anchored alternatives allocate 5–10–5 or 10–10–10 bins below, within, and above the reference interval, with equal-frequency binning within each region. Reference limits come from MIMIC-IV and differ from the severe abnormality thresholds used as prediction targets. Experiment 1 crosses these six quantization schemes with both layouts, yielding 12 configurations.

Numeric Encoding. Experiment 2 compares four encoders: discrete deciles, soft discretization, multiplicative xVal, and affine xVal. Discrete encoding assigns each value to a quantile-bin token. Soft discretization instead interpolates between adjacent quantile embeddings according to the value's position between their breakpoints, using the endpoint embedding outside the stored range (Norouzi et al., 2014).

We evaluate two code-normalized implementations of xVal, which encode continuous values by scaling a shared numeric embedding Golkar et al. (2023). Each numeric event is represented by a code token followed by a numeric placeholder. Values are centered by their code-specific training median, divided by the training interquartile range / 1.35 with a positive floor, and clipped to [-5, 5]. Multiplicative xVal scales the placeholder embedding by this normalized value; affine xVal additionally adds a learned bias. Both use an auxiliary numerical-prediction head. Appendix G provides formal definitions.

Temporal Encoding. We compare explicit time tokens, event order with unit-increment position identifiers, and admission-relative RoPE using elapsed time at 30-second resolution (Su et al., 2024). Event order and admission-relative RoPE omit time tokens. Crossing these three temporal encodings with the four numerical encodings yields 12 configurations in Experiment 2.

3.2. Native MIMIC versus CLIF Schema

Experiment 3 compares Native MIMIC MEDS streams with their CLIF 2.1.0 conversion (Rojas et al., 2025; Liao et al., 2026). Both pipelines use unfused deciles with explicit time tokens. CLIF inputs include admission, discharge, and transfer events; lab measures, vital signs, patient assessments, position, respiratory support, medication administration, continuous renal replacement therapy, and code status. Admissions and outcome labels are identical across pipelines (derived from native MIMIC). Event selection, code consolidation, and preprocessing vary between the arms, so this experiment evaluates the combined effect.

3.3. Training and Evaluation Setup

Models and Training. We train scaled-down causal decoders from scratch based on the Llama-3.2 (Grattafiori et al., 2024) and Qwen3 architectures (Yang et al., 2025). Each has 8 layers, 8 attention heads, hidden dimension 1024, intermediate dimension 2048, and a 4096-token context, yielding approximately 123.5M parameters for Llama and 103.7M parameters for Qwen. Qwen ties input and output embeddings while our implementation of Llama does not. We train using three seeds for each configuration, yielding 72, 72, and 12 runs for Experiments 1-3, respectively.

All runs use AdamW with learning rate (10^{-4}), weight decay 0.01, and minibatch size 4, for at most five epochs. Validation occurs twice per epoch; training stops after three consecutive evaluations without improvement. Experiment 1 uses packed sequences; Experiments 2–3 use admission-level windows. Discrete models optimize next-token cross-entropy. Soft discretization uses weighted targets for adjacent bins; xVal adds a numerical regression loss at numeric-placeholder positions. Token exposure, update counts, parameter counts, and computation vary across configurations despite the shared recipe. Additional training settings and parameter counts appear in Ap-

Table 1. Experiment 1: Effect of quantization on unfused tokenization. The reference column reports performance with equal-frequency deciles; all other columns show differences from this reference, with positive Δ indicating improvement. Equal-frequency quantiles use 20, 30, or 100 bins. Reference-range anchored quantiles allocate 5–10–5 or 10–10–10 bins below, within, and above laboratory reference ranges. Performance is averaged across outcomes within each family and then across two architectures and three seeds. * indicates Benjamini–Hochberg-adjusted $p < 0.05$ (Appendix I).

	Reference	Ventiles		Trentiles		Centiles
Family	Mean	Δ equal-frequency	Δ 5–10–5	Δ equal-frequency	Δ 10–10–10	Δ equal-frequency
Binary families, AUROC						
Hospital	0.884	+0.001	+0.001	+0.002	+0.000	+0.000
Organ support	0.958	+0.000	+0.000	+0.001	+0.001	+0.000
Laboratory	0.780	−0.006	−0.009	+0.002	−0.003	−0.011
Vital signs	0.694	+0.001	−0.001	+0.003	+0.001	+0.000
Regression families, Spearman ρ						
Electrolytes	0.450	−0.007*	−0.006*	−0.010*	−0.011*	−0.010*
Biomarkers	0.439	−0.001	−0.009*	−0.006	−0.004	−0.011*
Hematology	0.705	+0.002*	−0.001	−0.004*	−0.004*	−0.010*
Vital signs	0.465	+0.000	+0.001	+0.002	+0.000	−0.002

pendix E; secondary evaluation metrics are defined in Appendix D.

Classifier Probes and Metrics. We extract the final-layer hidden state at the last token preceding a pad, truncation, or termination token in the 24-hour input sequence. Last-token features are standardized with `StandardScaler` and evaluated with logistic regression (C=0.01) for binary outcomes and Ridge regression ($\alpha = 100$) for numeric outcomes (Table 4).

Binary outcomes are scored with area under the receiver operating characteristic curve (AUROC), Brier score, and expected calibration error with 15 bins; regression outcomes are scored with Spearman correlation, coefficient of determination, mean absolute error, and root mean squared error. Features are standardized using means and variances estimated from the probe-training split. Logistic regression uses a fixed inverse regularization hyperparameter of 0.01, and Ridge regression uses a fixed regularization hyperparameter of 100 (Appendix D). Main text tables report family-mean AUROC and Spearman correlation. For each trained model, outcomes within a family receive equal weight; the resulting family score is then averaged across three training seeds and two architectures. The families comprise hospital outcomes, organ support, laboratory extrema, and physiologic measurements, with their members listed in Table 4. Main tables report AUROC or Spearman correlation depending on outcome type. For each model, we average metrics across outcomes within the family and then average across the architectures and seeds. Appendix B provides full granular outcome-level results and shows variation across trained models.

Statistical Comparisons. We compare family-mean performance using two-sided paired prediction-swap tests with 1,000 permutations. Benjamini–Hochberg adjustment is applied separately within quantization, fusion, numerical encoding, temporal encoding, and schema comparisons; a star denotes an adjusted p-value below 0.05 (Benjamini and Hochberg, 1995). Tests operate at the admission level and condition on the six fitted models; they do not account for dependence between admissions from the same patient. Baselines, permutation procedures, and uncertainty summaries are detailed in Appendix I.

4. Results

Table 1–Table 3 summarize family-mean performance across the two architectures and three seeds. Appendix B provide granular outcome-level results for all experiments.

Table 2. Experiment 1: Effect of code–value fusion at matched granularity. Each column reports fused minus unfused tokenization with the same bin boundaries.

Family	Deciles	Ventiles	5–10–5	Trentiles	10–10–10	Centiles
Binary families, AUROC						
Hospital	+0.004*	+0.003*	+0.004*	+0.000	+0.004*	+0.002
Organ support	+0.002	+0.001	+0.001	+0.000	+0.001	+0.001
Laboratory	+0.011*	+0.023*	+0.017*	+0.009	+0.016*	+0.025*
Vital signs	+0.033*	+0.032*	+0.031*	+0.028*	+0.026*	+0.027*
Regression families, Spearman ρ						
Electrolytes	+0.069*	+0.072*	+0.062*	+0.066*	+0.069*	+0.063*
Biomarkers	+0.025*	+0.017*	+0.027*	+0.021*	+0.016*	+0.015*
Hematology	+0.099*	+0.090*	+0.099*	+0.093*	+0.088*	+0.091*
Vital signs	+0.114*	+0.111*	+0.111*	+0.106*	+0.108*	+0.107*

Table 3. Experiment 3: Effect of CLIF schema remapping.

Family	Native MIMIC	CLIF 2.1.0	Δ
Binary families, AUROC			
Hospital	0.877	0.886	+0.010*
Organ support	0.926	0.917	−0.009*
Laboratory	0.729	0.735	+0.006
Vital signs	0.711	0.743	+0.032*
Regression families, Spearman ρ			
Electrolytes	0.425	0.390	−0.035*
Biomarkers	0.417	0.423	+0.005
Hematology	0.635	0.663	+0.029*
Vital signs	0.494	0.593	+0.099*

4.1. Experiment 1: Quantization and Code–Value Fusion

Fused tokenization produced the largest and most consistent family-mean differences. At matched population deciles, fused and unfused tokenizers share every bin boundary: the unfused representation uses adjacent code and shared quantile-bin tokens, whereas the fused representation uses a single code-specific value token. Fusion raised the six-model point estimate for all eight families (Table 2), by +0.002 to +0.033 AUROC and +0.025 to +0.114 Spearman correlation. The largest member-level changes occurred for heart-rate extrema: minimum heart-rate correlation rose from 0.445 to 0.694, and maximum heart-rate correlation rose from 0.451 to 0.644.

Increasing the number of population-quantile bins beyond ten produced small, non-monotonic family-mean differences relative to unfused deciles (Table 1). Reference-range anchoring also produced mixed differences relative to population quantiles with the same number of bins. No unfused alternative improved all eight family summaries.

4.2. Experiment 2: Numeric and Temporal Encoding

Numerical encoding. Experiment 2 crosses four numeric encodings with three temporal encodings at unfused population deciles. With time tokens fixed, discrete and soft encodings differed by at most 0.0021 AUROC across binary families, while discrete exceeded soft by 0.004–0.011 Spearman correlation across regression families (Table 6). Both xVal parameterizations were lower than discrete encoding in every family: the differences were 0.050–0.057 AUROC and 0.078–0.162 Spearman correlation for code-normalized xVal, and 0.051–0.057 AUROC and 0.076–0.157 for affine xVal. Affine xVal remained close to code-normalized xVal and below the discrete and soft encodings in every family.

Temporal encoding with discrete values. With discrete numeric encoding fixed, event order and admission-relative RoPE both produced higher mean performance than explicit time tokens on all eight fam-

ily summaries (Table 7). Hospital-outcome AUROC was 0.890 with time tokens, 0.895 with event order, and 0.894 with admission-relative RoPE. Omitting time tokens reduced median first-24h sequence length from 93 to 83 tokens, approximately 11% (Figure 5).

4.3. Experiment 3: Native MIMIC versus CLIF Schema

Across the same 57,690 training admissions, the tokenized full-hospitalization training split contained 684.7 million native MIMIC tokens and 195.9 million CLIF tokens (28.6% of the native total). Schema remapping also reduced the tokenizer vocabulary from 17,752 to 397 entries. CLIF produced higher family-mean point estimates for six of eight families (Table 3), with differences of +0.005 to +0.099. Organ-support AUROC decreased by 0.009, and electrolyte Spearman correlation decreased by 0.035.

5. Discussion

Tokenization substantially affected downstream prediction performance under a shared training recipe for a variety of configurations. As a whole, the results support three practical takeaways: prioritize code-value fusion, do not assume elaborate encodings will improve performance, and consider whether expert knowledge and common data models enable the pipeline to focus on the meaningful information.

Code–value fusion. Fused tokens improved mean performance across all 8 outcome families, particularly for numerical predictions. This aligns with and generalizes the fused-token advantage reported by Guo et al. (2026) from pediatric binary tasks to adult hospital outcomes and regression outcomes. Finer quantization and anchoring to reference-ranges produced less consistent benefits. These findings support fused deciles as the reasonable starting configuration. We must acknowledge that fusion changes vocabulary size, sequence length and embedding capacity together so more work is required to pinpoint attribution for the performance gains.

Require evidence that additional encoding complexity improves performance . Soft discretization provided only a minor advantage of discrete bins, while both of the xVal implementations we tested underperformed the simpler methods. Adding an affine bias did not resolve the deficit, leaving normalization, the auxiliary numerical objective, and representation geometry as possible contributors. These results concern the tested implementations under a common recipe and do not establish a general disadvantage of continuous encoding. Event order and admission-relative RoPE shortened sequences without reducing family-mean performance relative to explicit time tokens, consistent with prior tokenization comparisons (Attrach et al., 2025). These results support starting with simple encodings but acknowledge longer horizon tasks outside of critical care may be more likely to benefit from sophisticated time encoding.

Consider whether common schemas or expert knowledge can reduce vocabulary while retaining high-signal information. The pipelines using CLIF-standardized data retained 28.6% of the native full-hospitalization training tokens and reduced the vocabulary from 17,752 to 397 entries, while producing higher point estimates for six of eight families (Table 3). This performance illustrates both: a.) the value of the effort of dozens of clinician-scientists who contributed expert knowledge to define CLIF and ensure that it includes the high signal data; and b.) the potential gains from a reduced vocabulary size in a domain with far less data availability than many other domains transformers have been applied to.

Limitations. Training 156 transformers is a computationally expensive exercise which places a hard limit on the number of ablations which could be performed. Additionally, this study relies on results from a single site, and while it considers multiple model architectures, we fixed the model size across experiments. Variations like the code–value fusion change multiple factors at the same time (e.g., sequence length, vocabulary size, and the size of the embedding) so we estimate the combined effects of these changes.

6. Conclusion

Tokenization is a consequential design choice for generative medical event models. Across 156 training runs, changes in how clinical events were encoded produced substantial differences in frozen-probe performance under a shared training recipe. Fused code–value deciles provided a strong starting configuration, particularly for numerical forecasting, while finer quantization, reference-range anchoring, and alternative numerical encoders offered no consistent advantage. Event order and admission-relative RoPE shortened sequences while maintaining higher family-mean point estimates than explicit time tokens. CLIF retained

28.6% of the native full-hospitalization training tokens, while raising six of eight family means, although outcome-specific losses make task-level evaluation essential. These findings provide practical guidance for model development and motivate testing whether the observed advantages extend across institutions, training scales, and generative tasks. Tokenization should therefore be evaluated alongside architecture and training choices when developing models of clinical trajectories. Multi-site validation and extensions such as fused soft discretization or code-conditioned continuous injection remain open. The tokenizer sets a ceiling on what next-token training can extract from an EHR sequence, and revising it is among the cheapest interventions available before scaling the model or the data.

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Appendix A. Outcome Definitions and Labeled Counts

Table 4: **Clinical prediction outcomes in the 24–48h evaluation window.** Experiments 1–2 include all 17 binary and 20 regression outcomes. Experiment 3 evaluates 16 binary and 20 regression outcomes on the ICU cohort, omitting ICU admission because every admission in that cohort has an ICU stay. Hospital and ICU length-of-stay regressions are omitted because their values are not confined to the follow-up window.

Outcome	Definition and source field(s)
Binary outcomes (17)	
Hospital mortality in hours 24–48	A `MEDS_DEATH` event or discharge disposition indicating death occurs in $(24, 48]$ hours after admission.
Still hospitalized at hour 48	`dischtime` occurs after hour 48.
First ICU admission in hours 24–48	The first linked `icu/icustays.intime` occurs in $(24, 48]$; omitted in Experiment 3 because every admission in that cohort has an ICU stay.
Still in ICU at hour 48	A linked ICU interval contains hour 48.
Invasive mechanical ventilation	First invasive mechanical ventilation procedure (itemid 224385 or 225792) occurs in $(24, 48]$.
Vasopressor initiation	First qualifying vasopressor infusion occurs in $(24, 48]$.
Continuous renal replacement therapy initiation	First continuous renal replacement therapy event (chartevent 227290 or procedure 225802) occurs in $(24, 48]$.
Hemodialysis initiation	First hemodialysis event (chartevent 226499 or procedure 225441) occurs in $(24, 48]$.
Hyperkalemia	Maximum potassium in $(24, 48]$ is $\geq$6.5 mEq/L.
Severe hypokalemia	Minimum potassium in $(24, 48]$ is $<$2.5 mEq/L.
Profound hyponatremia	Minimum sodium in $(24, 48]$ is $<$120 mEq/L.
Severe hypernatremia	Maximum sodium in $(24, 48]$ is $\geq$160 mEq/L.
Hypoglycemia	Minimum glucose in $(24, 48]$ is $<$54 mg/dL.
Severe anemia	Minimum hemoglobin in $(24, 48]$ is $<$7 g/dL.
Tachycardia	Maximum heart rate in $(24, 48]$ is $\geq$130 bpm.
Severe hypertension	Maximum systolic blood pressure is $\geq$180 mmHg or maximum diastolic blood pressure is $\geq$120 mmHg in $(24, 48]$.
Hypotension	Minimum mean arterial pressure is $<$65 mmHg or minimum systolic blood pressure is $<$90 mmHg in $(24, 48]$.
Regression outcomes (20)	
Min/max creatinine	Extrema in $(24, 48]$ from laboratory itemid 50912.
Min/max troponin	Extrema in $(24, 48]$ from troponin laboratory events.
Min/max hemoglobin	Extrema in $(24, 48]$ from laboratory itemids 51222 and 50811.
Min/max potassium	Extrema in $(24, 48]$ from laboratory itemids 50971 and 50822.
Min/max glucose	Extrema in $(24, 48]$ from laboratory itemids 50931 and 50809.
Min/max sodium	Extrema in $(24, 48]$ from laboratory itemids 50983, 50824, and 52623.
Min/max B-type natriuretic peptide	Extrema in $(24, 48]$ from B-type and N-terminal pro-B-type natriuretic peptide laboratory events.
Min/max heart rate	Extrema in $(24, 48]$ from ICU heart-rate events.
Min/max systolic blood pressure	Extrema in $(24, 48]$ from ICU systolic blood-pressure events.
Min/max diastolic blood pressure	Extrema in $(24, 48]$ from ICU diastolic blood-pressure events.

Eligibility and leakage control. For intervention and threshold outcomes, admissions that already met the endpoint during the first 24 hours are excluded. Physiologic threshold and regression outcomes additionally require a qualifying measurement in hours 24–48. Absence of a qualifying intervention during the follow-up window is treated as negative. The resulting labeled counts are shown in Table 5.

Table 5. **Labeled test-set counts for 24–48h binary outcomes.** Experiments 1–2 use the hospitalization cohort (85,057 test hospitalizations). Experiment 3 uses the ICU cohort (16,529 test hospitalizations) and omits ICU admission because every hospitalization in that cohort has an ICU stay. Positives are events observed in hours 24–48 among hospitalizations that were eligible for that label.

Outcome	Exp1–2 labeled N	Exp1–2 pos / neg	Exp3 labeled N	Exp3 pos / neg
Hospital mortality in hours 24–48	85,057	238 / 84,819	16,529	209 / 16,320
Still hospitalized at hour 48	85,057	66,238 / 18,819	16,529	15,438 / 1,091
First ICU admission in hours 24–48	71,949	1,004 / 70,945	—	—
Still in ICU at hour 48	85,057	8,018 / 77,039	16,529	8,041 / 8,488
Invasive mechanical ventilation	83,278	1,153 / 82,125	14,766	1,224 / 13,542
Vasopressor initiation	82,207	592 / 81,615	13,654	642 / 13,012
CRRT initiation	84,947	56 / 84,891	16,429	71 / 16,358
Hemodialysis initiation	84,866	118 / 84,748	16,330	135 / 16,195
Hyperkalemia	58,971	158 / 58,813	14,635	99 / 14,536
Severe hypokalemia	59,656	36 / 59,620	14,984	26 / 14,958
Profound hyponatremia	59,350	29 / 59,321	14,858	17 / 14,841
Severe hypernatremia	59,502	16 / 59,486	14,976	16 / 14,960
Hypoglycemia	58,499	325 / 58,174	14,803	85 / 14,718
Severe anemia	57,619	888 / 56,731	13,991	341 / 13,650
Tachycardia	10,955	525 / 10,430	11,148	526 / 10,622
Severe hypertension	10,580	783 / 9,797	10,740	784 / 9,956
Hypotension	5,003	1,863 / 3,140	5,176	1,997 / 3,179

Table 6. Experiment 2: numeric encodings with time tokens. Δ versus discrete bins. Unsigned values are family means; signed values are Δ. * denotes BH FDR < 0.05.

	Reference	Soft discretization		Code-normalized xVal		Affine xVal	
Family	Discrete	Mean	Δ	Mean	Δ	Mean	Δ
Binary families, AUROC							
Hospital	0.890	0.889	−0.001	0.839	−0.050*	0.838	−0.052*
Organ support	0.959	0.959	−0.001	0.902	−0.057*	0.902	−0.057*
Laboratory	0.785	0.783	−0.002	0.729	−0.056*	0.734	−0.051*
Vital signs	0.701	0.701	+0.000	0.647	−0.054*	0.648	−0.053*
Regression families, Spearman ρ							
Electrolytes	0.471	0.460	−0.011*	0.346	−0.125*	0.360	−0.112*
Biomarkers	0.454	0.447	−0.008*	0.377	−0.078*	0.378	−0.076*
Hematology	0.731	0.723	−0.007*	0.568	−0.162*	0.574	−0.157*
Vital signs	0.473	0.469	−0.004*	0.379	−0.094*	0.377	−0.096*

Appendix B. Family-mean and Outcome-level Summaries

For each trained model, a family score is the equal-weight mean of its member outcomes. The family tables report the mean across three training seeds and two architectures, with the corresponding min/max in brackets. Outcome-level tables list the same six-model mean, min, and max in separate columns for every family member. These summaries describe variation across trained models. Run- and outcome-specific 95% percentile confidence intervals quantify test-set sampling variability conditional on each fitted probe; they were calculated from 1,000 bootstraps using random seed 42 and are stored in `outputs/runs/ml4h_2026/metrics/metrics_long_with_ci.parquet` in the `input-representation-benchmark` repository. Family-mean Δ tests use the paired admission-swap procedure in Appendix I. The resulting p-values and Benjamini–Hochberg decisions are stored in `outputs/runs/ml4h_2026/family_effect_bootstrap/family_effect_intervals.parquet`.

Table 7. Experiment 2: temporal encodings with discrete values. Δ versus time tokens.

	Reference	Event order		Hospitalization-relative RoPE	
Family	Time tokens	Mean	Δ	Mean	Δ
Binary families, AUROC					
Hospital	0.890	0.895	+0.005*	0.894	+0.005*
Organ support	0.959	0.963	+0.003*	0.961	+0.002
Laboratory	0.785	0.794	+0.009	0.793	+0.008
Vital signs	0.701	0.707	+0.006*	0.708	+0.007*
Regression families, Spearman ρ					
Electrolytes	0.471	0.490	+0.019*	0.491	+0.019*
Biomarkers	0.454	0.466	+0.012*	0.456	+0.002
Hematology	0.731	0.744	+0.014*	0.749	+0.019*
Vital signs	0.473	0.487	+0.014*	0.491	+0.018*

Table 8. Family-mean AUROC and Spearman ρ for every Experiment 1 configuration. Each cell reports the mean across six trained models (three seeds × two architectures), with min/max in brackets. Member outcomes receive equal weight within each model before the six-model summary.

Configuration	Hospital AUROC	Organ-support AUROC	Laboratory AUROC	Vital-sign AUROC	Electrolyte ρ	Biomarker ρ	Hematology ρ	Vital-sign ρ
Deciles · population, unfused	0.884 [0.882, 0.887] 0.888	0.958 [0.955, 0.959] 0.960	0.780 [0.774, 0.785] 0.791	0.694 [0.690, 0.702] 0.727	0.450 [0.440, 0.478] 0.519	0.439 [0.428, 0.453] 0.464	0.705 [0.696, 0.712] 0.804	0.465 [0.461, 0.471] 0.578
Deciles · population, fused	[0.886, 0.891] 0.886	[0.959, 0.961] 0.957	[0.775, 0.800] 0.774	[0.723, 0.733] 0.696	[0.513, 0.530] 0.443	[0.453, 0.473] 0.438	[0.799, 0.813] 0.707	[0.571, 0.583] 0.465
Ventiles · population, unfused	[0.884, 0.888] 0.888	[0.955, 0.961] 0.958	[0.764, 0.791] 0.797	[0.689, 0.701] 0.728	[0.436, 0.447] 0.515	[0.430, 0.448] 0.455	[0.697, 0.718] 0.797	[0.459, 0.473] 0.576
Ventiles · population, fused	[0.886, 0.890] 0.885	[0.951, 0.960] 0.958	[0.789, 0.805] 0.771	[0.722, 0.733] 0.694	[0.504, 0.527] 0.444	[0.449, 0.462] 0.430	[0.789, 0.801] 0.704	[0.567, 0.583] 0.466
Ventiles · 5-10-5 reference, unfused	[0.884, 0.888] 0.889	[0.956, 0.960] 0.959	[0.766, 0.780] 0.788	[0.688, 0.703] 0.725	[0.422, 0.461] 0.506	[0.417, 0.442] 0.457	[0.694, 0.722] 0.804	[0.455, 0.473] 0.577
Ventiles · 5-10-5 reference, fused	[0.887, 0.890] 0.886	[0.957, 0.960] 0.959	[0.780, 0.796] 0.782	[0.721, 0.727] 0.697	[0.495, 0.518] 0.440	[0.450, 0.466] 0.433	[0.801, 0.806] 0.701	[0.570, 0.580] 0.467
Trentiles · population, unfused	[0.884, 0.888] 0.886	[0.957, 0.961] 0.959	[0.763, 0.794] 0.791	[0.694, 0.702] 0.725	[0.429, 0.458] 0.506	[0.424, 0.447] 0.454	[0.693, 0.711] 0.794	[0.459, 0.473] 0.573
Trentiles · population, fused	[0.885, 0.887] 0.884	[0.958, 0.960] 0.958	[0.783, 0.803] 0.777	[0.718, 0.732] 0.695	[0.488, 0.517] 0.439	[0.445, 0.465] 0.435	[0.781, 0.807] 0.701	[0.566, 0.576] 0.465
Trentiles · 10-10-10 reference, unfused	[0.883, 0.886] 0.888	[0.955, 0.960] 0.959	[0.772, 0.784] 0.793	[0.695, 0.696] 0.722	[0.436, 0.446] 0.509	[0.430, 0.439] 0.451	[0.690, 0.709] 0.789	[0.457, 0.476] 0.573
Trentiles · 10-10-10 reference, fused	[0.885, 0.890] 0.884	[0.958, 0.959] 0.958	[0.772, 0.804] 0.769	[0.716, 0.727] 0.695	[0.500, 0.512] 0.440	[0.444, 0.458] 0.428	[0.768, 0.807] 0.695	[0.569, 0.576] 0.463
Centiles · population, unfused	[0.879, 0.888] 0.886	[0.954, 0.961] 0.959	[0.762, 0.777] 0.794	[0.692, 0.698] 0.722	[0.423, 0.458] 0.504	[0.421, 0.438] 0.443	[0.686, 0.703] 0.786	[0.455, 0.470] 0.570
Centiles · population, fused	[0.883, 0.890]	[0.956, 0.961]	[0.780, 0.811]	[0.720, 0.727]	[0.491, 0.521]	[0.428, 0.454]	[0.772, 0.801]	[0.565, 0.575]

Table 9: Outcome-level AUROC and Spearman ρ for every Experiment 1 configuration. Mean, Min, and Max summarize six trained models (three seeds × two architectures).

Outcome	Configuration	Mean	Min	Max
Hospital outcomes (AUROC)				
Hospital mortality in hours 24–48	Deciles · population, unfused	0.958	0.951	0.963
	Deciles · population, fused	0.964	0.960	0.971
	Ventiles · population, unfused	0.958	0.954	0.962
	Ventiles · population, fused	0.964	0.961	0.968
	Ventiles · 5-10-5 reference, unfused	0.961	0.957	0.963

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Ventiles · 5-10-5 reference, fused	0.966	0.962	0.969
	Trentiles · population, unfused	0.962	0.959	0.966
	Trentiles · population, fused	0.963	0.960	0.965
	Trentiles · 10-10-10 reference, unfused	0.957	0.952	0.959
	Trentiles · 10-10-10 reference, fused	0.965	0.954	0.970
	Centiles · population, unfused	0.958	0.952	0.964
	Centiles · population, fused	0.963	0.960	0.969
Still hospitalized at hour 48	Deciles · population, unfused	0.803	0.800	0.806
	Deciles · population, fused	0.808	0.803	0.810
	Ventiles · population, unfused	0.805	0.803	0.807
	Ventiles · population, fused	0.808	0.807	0.809
	Ventiles · 5-10-5 reference, unfused	0.804	0.800	0.808
	Ventiles · 5-10-5 reference, fused	0.809	0.806	0.811
	Trentiles · population, unfused	0.805	0.802	0.806
	Trentiles · population, fused	0.805	0.802	0.808
	Trentiles · 10-10-10 reference, unfused	0.805	0.804	0.807
	Trentiles · 10-10-10 reference, fused	0.806	0.805	0.809
	Centiles · population, unfused	0.804	0.798	0.809
	Centiles · population, fused	0.804	0.800	0.807
First ICU admission in hours 24–48	Deciles · population, unfused	0.820	0.816	0.825
	Deciles · population, fused	0.826	0.819	0.831
	Ventiles · population, unfused	0.823	0.816	0.828
	Ventiles · population, fused	0.825	0.818	0.829
	Ventiles · 5-10-5 reference, unfused	0.822	0.816	0.828
	Ventiles · 5-10-5 reference, fused	0.825	0.822	0.828
	Trentiles · population, unfused	0.823	0.819	0.827
	Trentiles · population, fused	0.823	0.820	0.827
	Trentiles · 10-10-10 reference, unfused	0.819	0.815	0.826
	Trentiles · 10-10-10 reference, fused	0.824	0.819	0.827
	Centiles · population, unfused	0.820	0.813	0.825
	Centiles · population, fused	0.822	0.815	0.831
Still in ICU at hour 48	Deciles · population, unfused	0.955	0.955	0.955
	Deciles · population, fused	0.956	0.956	0.957
	Ventiles · population, unfused	0.955	0.955	0.956
	Ventiles · population, fused	0.956	0.955	0.957
	Ventiles · 5-10-5 reference, unfused	0.955	0.954	0.955
	Ventiles · 5-10-5 reference, fused	0.956	0.955	0.956
	Trentiles · population, unfused	0.955	0.954	0.956
	Trentiles · population, fused	0.955	0.954	0.956
	Trentiles · 10-10-10 reference, unfused	0.955	0.954	0.956

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Trentiles · 10-10-10 reference, fused	0.956	0.955	0.956
	Centiles · population, unfused	0.955	0.953	0.956
	Centiles · population, fused	0.956	0.955	0.957
Organ support (AUROC)				
Invasive mechanical ventilation	Deciles · population, unfused	0.956	0.953	0.958
	Deciles · population, fused	0.958	0.956	0.959
	Ventiles · population, unfused	0.956	0.955	0.957
	Ventiles · population, fused	0.957	0.955	0.960
	Ventiles · 5-10-5 reference, unfused	0.956	0.955	0.958
	Ventiles · 5-10-5 reference, fused	0.959	0.957	0.962
	Trentiles · population, unfused	0.956	0.954	0.959
	Trentiles · population, fused	0.957	0.955	0.958
	Trentiles · 10-10-10 reference, unfused	0.956	0.955	0.957
	Trentiles · 10-10-10 reference, fused	0.958	0.957	0.959
	Centiles · population, unfused	0.955	0.954	0.957
	Centiles · population, fused	0.956	0.954	0.960
Vasopressor initiation	Deciles · population, unfused	0.907	0.903	0.909
	Deciles · population, fused	0.909	0.907	0.912
	Ventiles · population, unfused	0.906	0.902	0.909
	Ventiles · population, fused	0.908	0.897	0.912
	Ventiles · 5-10-5 reference, unfused	0.908	0.905	0.912
	Ventiles · 5-10-5 reference, fused	0.909	0.905	0.911
	Trentiles · population, unfused	0.909	0.906	0.915
	Trentiles · population, fused	0.907	0.903	0.913
	Trentiles · 10-10-10 reference, unfused	0.908	0.905	0.911
	Trentiles · 10-10-10 reference, fused	0.910	0.906	0.915
	Centiles · population, unfused	0.909	0.905	0.911
	Centiles · population, fused	0.910	0.905	0.914
Continuous renal replacement	Deciles · population, unfused	0.987	0.979	0.992
	Deciles · population, fused	0.987	0.982	0.993
	Ventiles · population, unfused	0.985	0.978	0.994
	Ventiles · population, fused	0.984	0.970	0.989
	Ventiles · 5-10-5 reference, unfused	0.985	0.976	0.992
	Ventiles · 5-10-5 reference, fused	0.984	0.980	0.987
	Trentiles · population, unfused	0.987	0.978	0.992
	Trentiles · population, fused	0.988	0.983	0.993
	Trentiles · 10-10-10 reference, unfused	0.989	0.986	0.991
	Trentiles · 10-10-10 reference, fused	0.986	0.982	0.989
	Centiles · population, unfused	0.986	0.974	0.991
	Centiles · population, fused	0.987	0.983	0.993
Hemodialysis initiation	Deciles · population, unfused	0.982	0.976	0.986

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Deciles · population, fused	0.986	0.982	0.988
	Ventiles · population, unfused	0.983	0.981	0.985
	Ventiles · population, fused	0.983	0.980	0.987
	Ventiles · 5-10-5 reference, unfused	0.983	0.978	0.989
	Ventiles · 5-10-5 reference, fused	0.983	0.979	0.988
	Trentiles · population, unfused	0.983	0.979	0.987
	Trentiles · population, fused	0.984	0.980	0.988
	Trentiles · 10-10-10 reference, unfused	0.980	0.970	0.983
	Trentiles · 10-10-10 reference, fused	0.982	0.976	0.985
	Centiles · population, unfused	0.982	0.979	0.987
	Centiles · population, fused	0.982	0.977	0.986
Laboratory (AUROC)				
Hyperkalemia	Deciles · population, unfused	0.734	0.720	0.752
	Deciles · population, fused	0.711	0.698	0.724
	Ventiles · population, unfused	0.719	0.706	0.752
	Ventiles · population, fused	0.709	0.697	0.733
	Ventiles · 5-10-5 reference, unfused	0.723	0.712	0.733
	Ventiles · 5-10-5 reference, fused	0.713	0.687	0.743
	Trentiles · population, unfused	0.720	0.711	0.732
	Trentiles · population, fused	0.722	0.703	0.736
	Trentiles · 10-10-10 reference, unfused	0.727	0.701	0.749
	Trentiles · 10-10-10 reference, fused	0.712	0.681	0.738
	Centiles · population, unfused	0.722	0.703	0.750
	Centiles · population, fused	0.720	0.689	0.735
Severe hypokalemia	Deciles · population, unfused	0.690	0.665	0.703
	Deciles · population, fused	0.693	0.619	0.753
	Ventiles · population, unfused	0.678	0.634	0.735
	Ventiles · population, fused	0.720	0.685	0.748
	Ventiles · 5-10-5 reference, unfused	0.685	0.663	0.729
	Ventiles · 5-10-5 reference, fused	0.684	0.652	0.740
	Trentiles · population, unfused	0.702	0.647	0.734
	Trentiles · population, fused	0.668	0.628	0.726
	Trentiles · 10-10-10 reference, unfused	0.671	0.643	0.725
	Trentiles · 10-10-10 reference, fused	0.699	0.644	0.741
	Centiles · population, unfused	0.676	0.609	0.717
	Centiles · population, fused	0.697	0.645	0.745
Severe anemia	Deciles · population, unfused	0.846	0.842	0.850
	Deciles · population, fused	0.889	0.886	0.894
	Ventiles · population, unfused	0.847	0.843	0.857
	Ventiles · population, fused	0.883	0.877	0.889
	Ventiles · 5-10-5 reference, unfused	0.848	0.840	0.857
	Ventiles · 5-10-5 reference, fused	0.888	0.882	0.892
	Trentiles · population, unfused	0.844	0.838	0.850
	Trentiles · population, fused	0.882	0.878	0.886

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Trentiles · 10-10-10 reference, unfused	0.847	0.840	0.858
	Trentiles · 10-10-10 reference, fused	0.882	0.872	0.887
	Centiles · population, unfused	0.840	0.834	0.847
	Centiles · population, fused	0.878	0.869	0.884
Hypoglycemia	Deciles · population, unfused	0.721	0.707	0.734
	Deciles · population, fused	0.748	0.741	0.758
	Ventiles · population, unfused	0.717	0.706	0.737
	Ventiles · population, fused	0.742	0.732	0.751
	Ventiles · 5-10-5 reference, unfused	0.717	0.705	0.732
	Ventiles · 5-10-5 reference, fused	0.745	0.730	0.757
	Trentiles · population, unfused	0.717	0.708	0.739
	Trentiles · population, fused	0.743	0.730	0.756
	Trentiles · 10-10-10 reference, unfused	0.723	0.698	0.738
	Trentiles · 10-10-10 reference, fused	0.737	0.724	0.747
	Centiles · population, unfused	0.721	0.709	0.732
	Centiles · population, fused	0.738	0.726	0.751
Profound hyponatremia	Deciles · population, unfused	0.740	0.727	0.759
	Deciles · population, fused	0.745	0.720	0.785
	Ventiles · population, unfused	0.736	0.690	0.792
	Ventiles · population, fused	0.767	0.741	0.781
	Ventiles · 5-10-5 reference, unfused	0.701	0.648	0.747
	Ventiles · 5-10-5 reference, fused	0.744	0.697	0.820
	Trentiles · population, unfused	0.752	0.695	0.802
	Trentiles · population, fused	0.768	0.739	0.813
	Trentiles · 10-10-10 reference, unfused	0.742	0.725	0.767
	Trentiles · 10-10-10 reference, fused	0.770	0.713	0.805
	Centiles · population, unfused	0.718	0.667	0.774
	Centiles · population, fused	0.783	0.775	0.793
Severe hypernatremia	Deciles · population, unfused	0.948	0.933	0.963
	Deciles · population, fused	0.960	0.930	0.983
	Ventiles · population, unfused	0.947	0.920	0.968
	Ventiles · population, fused	0.959	0.941	0.982
	Ventiles · 5-10-5 reference, unfused	0.953	0.894	0.987
	Ventiles · 5-10-5 reference, fused	0.956	0.909	0.980
	Trentiles · population, unfused	0.955	0.937	0.974
	Trentiles · population, fused	0.965	0.922	0.987
	Trentiles · 10-10-10 reference, unfused	0.953	0.933	0.981
	Trentiles · 10-10-10 reference, fused	0.959	0.927	0.974
	Centiles · population, unfused	0.938	0.881	0.977
	Centiles · population, fused	0.951	0.911	0.976
Vitals (AUROC)				
Tachycardia	Deciles · population, unfused	0.692	0.684	0.699

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Deciles · population, fused	0.738	0.727	0.747
	Ventiles · population, unfused	0.695	0.681	0.708
	Ventiles · population, fused	0.746	0.731	0.755
	Ventiles · 5-10-5 reference, unfused	0.693	0.680	0.709
	Ventiles · 5-10-5 reference, fused	0.736	0.724	0.748
	Trentiles · population, unfused	0.695	0.684	0.711
	Trentiles · population, fused	0.735	0.722	0.749
	Trentiles · 10-10-10 reference, unfused	0.692	0.683	0.701
	Trentiles · 10-10-10 reference, fused	0.738	0.726	0.747
	Centiles · population, unfused	0.691	0.683	0.695
	Centiles · population, fused	0.737	0.727	0.746
Severe hypertension	Deciles · population, unfused	0.648	0.633	0.660
	Deciles · population, fused	0.688	0.682	0.698
	Ventiles · population, unfused	0.650	0.642	0.657
	Ventiles · population, fused	0.685	0.679	0.691
	Ventiles · 5-10-5 reference, unfused	0.648	0.633	0.659
	Ventiles · 5-10-5 reference, fused	0.685	0.673	0.694
	Trentiles · population, unfused	0.651	0.647	0.657
	Trentiles · population, fused	0.690	0.680	0.696
	Trentiles · 10-10-10 reference, unfused	0.650	0.644	0.658
	Trentiles · 10-10-10 reference, fused	0.679	0.673	0.687
	Centiles · population, unfused	0.652	0.644	0.656
	Centiles · population, fused	0.680	0.673	0.689
Hypotension	Deciles · population, unfused	0.743	0.736	0.753
	Deciles · population, fused	0.755	0.749	0.760
	Ventiles · population, unfused	0.743	0.732	0.749
	Ventiles · population, fused	0.752	0.750	0.755
	Ventiles · 5-10-5 reference, unfused	0.740	0.735	0.745
	Ventiles · 5-10-5 reference, fused	0.752	0.748	0.754
	Trentiles · population, unfused	0.745	0.742	0.748
	Trentiles · population, fused	0.750	0.746	0.754
	Trentiles · 10-10-10 reference, unfused	0.743	0.737	0.748
	Trentiles · 10-10-10 reference, fused	0.748	0.746	0.753
	Centiles · population, unfused	0.741	0.737	0.744
	Centiles · population, fused	0.749	0.746	0.753
Electrolytes (Spearman ρ)				
Minimum potassium	Deciles · population, unfused	0.472	0.459	0.493
	Deciles · population, fused	0.505	0.493	0.519
	Ventiles · population, unfused	0.457	0.445	0.466
	Ventiles · population, fused	0.497	0.486	0.510
	Ventiles · 5-10-5 reference, unfused	0.466	0.455	0.478
	Ventiles · 5-10-5 reference, fused	0.493	0.478	0.503
	Trentiles · population, unfused	0.464	0.455	0.475
	Trentiles · population, fused	0.498	0.481	0.508

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Trentiles · 10-10-10 reference, unfused	0.463	0.458	0.468
	Trentiles · 10-10-10 reference, fused	0.496	0.483	0.510
	Centiles · population, unfused	0.458	0.451	0.470
	Centiles · population, fused	0.492	0.478	0.513
Peak potassium	Deciles · population, unfused	0.468	0.455	0.490
	Deciles · population, fused	0.502	0.491	0.514
	Ventiles · population, unfused	0.455	0.445	0.466
	Ventiles · population, fused	0.493	0.482	0.506
	Ventiles · 5-10-5 reference, unfused	0.462	0.451	0.474
	Ventiles · 5-10-5 reference, fused	0.488	0.471	0.497
	Trentiles · population, unfused	0.460	0.449	0.470
	Trentiles · population, fused	0.492	0.476	0.501
	Trentiles · 10-10-10 reference, unfused	0.459	0.455	0.468
	Trentiles · 10-10-10 reference, fused	0.491	0.477	0.505
	Centiles · population, unfused	0.454	0.446	0.466
	Centiles · population, fused	0.488	0.474	0.507
Minimum sodium	Deciles · population, unfused	0.432	0.415	0.466
	Deciles · population, fused	0.535	0.527	0.544
	Ventiles · population, unfused	0.431	0.417	0.453
	Ventiles · population, fused	0.536	0.523	0.553
	Ventiles · 5-10-5 reference, unfused	0.426	0.392	0.449
	Ventiles · 5-10-5 reference, fused	0.523	0.509	0.540
	Trentiles · population, unfused	0.420	0.405	0.444
	Trentiles · population, fused	0.518	0.499	0.538
	Trentiles · 10-10-10 reference, unfused	0.420	0.414	0.438
	Trentiles · 10-10-10 reference, fused	0.524	0.510	0.530
	Centiles · population, unfused	0.426	0.394	0.466
	Centiles · population, fused	0.518	0.498	0.533
Maximum sodium	Deciles · population, unfused	0.427	0.409	0.462
	Deciles · population, fused	0.534	0.524	0.543
	Ventiles · population, unfused	0.427	0.414	0.446
	Ventiles · population, fused	0.535	0.523	0.553
	Ventiles · 5-10-5 reference, unfused	0.422	0.389	0.446
	Ventiles · 5-10-5 reference, fused	0.522	0.510	0.537
	Trentiles · population, unfused	0.416	0.399	0.441
	Trentiles · population, fused	0.517	0.497	0.539
	Trentiles · 10-10-10 reference, unfused	0.416	0.408	0.433
	Trentiles · 10-10-10 reference, fused	0.523	0.510	0.529
	Centiles · population, unfused	0.422	0.389	0.462
	Centiles · population, fused	0.516	0.496	0.530
Biomarkers (Spearman ρ)				
Minimum creatinine	Deciles · population, unfused	0.556	0.548	0.563
	Deciles · population, fused	0.576	0.571	0.588

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Ventiles · population, unfused	0.549	0.540	0.566
	Ventiles · population, fused	0.562	0.543	0.572
	Ventiles · 5-10-5 reference, unfused	0.545	0.531	0.567
	Ventiles · 5-10-5 reference, fused	0.561	0.552	0.565
	Trentiles · population, unfused	0.542	0.530	0.561
	Trentiles · population, fused	0.559	0.541	0.581
	Trentiles · 10-10-10 reference, unfused	0.543	0.528	0.553
	Trentiles · 10-10-10 reference, fused	0.557	0.551	0.565
	Centiles · population, unfused	0.536	0.530	0.549
	Centiles · population, fused	0.550	0.537	0.569
Peak creatinine	Deciles · population, unfused	0.560	0.553	0.567
	Deciles · population, fused	0.578	0.574	0.589
	Ventiles · population, unfused	0.553	0.546	0.570
	Ventiles · population, fused	0.565	0.546	0.576
	Ventiles · 5-10-5 reference, unfused	0.549	0.536	0.571
	Ventiles · 5-10-5 reference, fused	0.563	0.555	0.568
	Trentiles · population, unfused	0.546	0.534	0.565
	Trentiles · population, fused	0.561	0.544	0.583
	Trentiles · 10-10-10 reference, unfused	0.547	0.533	0.557
	Trentiles · 10-10-10 reference, fused	0.560	0.554	0.568
	Centiles · population, unfused	0.541	0.535	0.552
	Centiles · population, fused	0.553	0.540	0.571
Minimum troponin	Deciles · population, unfused	0.455	0.441	0.463
	Deciles · population, fused	0.471	0.460	0.484
	Ventiles · population, unfused	0.438	0.418	0.461
	Ventiles · population, fused	0.459	0.438	0.486
	Ventiles · 5-10-5 reference, unfused	0.428	0.415	0.435
	Ventiles · 5-10-5 reference, fused	0.480	0.470	0.499
	Trentiles · population, unfused	0.439	0.423	0.460
	Trentiles · population, fused	0.469	0.445	0.487
	Trentiles · 10-10-10 reference, unfused	0.440	0.429	0.466
	Trentiles · 10-10-10 reference, fused	0.461	0.439	0.486
	Centiles · population, unfused	0.444	0.432	0.462
	Centiles · population, fused	0.457	0.437	0.487
Peak troponin	Deciles · population, unfused	0.442	0.430	0.450
	Deciles · population, fused	0.461	0.448	0.468
	Ventiles · population, unfused	0.427	0.406	0.448
	Ventiles · population, fused	0.448	0.423	0.477
	Ventiles · 5-10-5 reference, unfused	0.420	0.404	0.426
	Ventiles · 5-10-5 reference, fused	0.470	0.461	0.483
	Trentiles · population, unfused	0.429	0.419	0.448
	Trentiles · population, fused	0.465	0.448	0.475
	Trentiles · 10-10-10 reference, unfused	0.429	0.414	0.456

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Trentiles · 10-10-10 reference, fused	0.452	0.431	0.479
	Centiles · population, unfused	0.435	0.423	0.455
	Centiles · population, fused	0.453	0.429	0.478
Minimum glucose	Deciles · population, unfused	0.443	0.435	0.451
	Deciles · population, fused	0.474	0.471	0.479
	Ventiles · population, unfused	0.441	0.434	0.451
	Ventiles · population, fused	0.467	0.462	0.469
	Ventiles · 5-10-5 reference, unfused	0.441	0.433	0.448
	Ventiles · 5-10-5 reference, fused	0.471	0.465	0.478
	Trentiles · population, unfused	0.440	0.433	0.446
	Trentiles · population, fused	0.467	0.460	0.478
	Trentiles · 10-10-10 reference, unfused	0.442	0.431	0.448
	Trentiles · 10-10-10 reference, fused	0.467	0.464	0.474
	Centiles · population, unfused	0.432	0.426	0.439
	Centiles · population, fused	0.456	0.444	0.469
Maximum glucose	Deciles · population, unfused	0.485	0.480	0.490
	Deciles · population, fused	0.512	0.508	0.514
	Ventiles · population, unfused	0.486	0.479	0.494
	Ventiles · population, fused	0.507	0.502	0.511
	Ventiles · 5-10-5 reference, unfused	0.484	0.478	0.492
	Ventiles · 5-10-5 reference, fused	0.509	0.505	0.516
	Trentiles · population, unfused	0.485	0.479	0.490
	Trentiles · population, fused	0.505	0.499	0.515
	Trentiles · 10-10-10 reference, unfused	0.486	0.478	0.491
	Trentiles · 10-10-10 reference, fused	0.506	0.502	0.511
	Centiles · population, unfused	0.479	0.473	0.483
	Centiles · population, fused	0.497	0.484	0.509
Minimum natriuretic peptide	Deciles · population, unfused	0.289	0.251	0.333
	Deciles · population, fused	0.322	0.292	0.349
	Ventiles · population, unfused	0.309	0.278	0.334
	Ventiles · population, fused	0.319	0.299	0.344
	Ventiles · 5-10-5 reference, unfused	0.287	0.227	0.318
	Ventiles · 5-10-5 reference, fused	0.303	0.284	0.331
	Trentiles · population, unfused	0.294	0.264	0.347
	Trentiles · population, fused	0.308	0.282	0.355
	Trentiles · 10-10-10 reference, unfused	0.298	0.285	0.309
	Trentiles · 10-10-10 reference, fused	0.303	0.271	0.333
	Centiles · population, unfused	0.280	0.239	0.308
	Centiles · population, fused	0.291	0.232	0.328
Peak natriuretic peptide	Deciles · population, unfused	0.282	0.237	0.329
	Deciles · population, fused	0.317	0.294	0.349
	Ventiles · population, unfused	0.301	0.275	0.325

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Ventiles · population, fused	0.313	0.287	0.340
	Ventiles · 5-10-5 reference, unfused	0.285	0.224	0.323
	Ventiles · 5-10-5 reference, fused	0.301	0.277	0.321
	Trentiles · population, unfused	0.286	0.259	0.337
	Trentiles · population, fused	0.299	0.274	0.342
	Trentiles · 10-10-10 reference, unfused	0.297	0.285	0.310
	Trentiles · 10-10-10 reference, fused	0.301	0.267	0.328
	Centiles · population, unfused	0.278	0.235	0.305
	Centiles · population, fused	0.285	0.228	0.322
Hematology (Spearman ρ)				
Minimum hemoglobin	Deciles · population, unfused	0.708	0.698	0.715
	Deciles · population, fused	0.804	0.798	0.812
	Ventiles · population, unfused	0.710	0.700	0.721
	Ventiles · population, fused	0.797	0.789	0.801
	Ventiles · 5-10-5 reference, unfused	0.707	0.697	0.725
	Ventiles · 5-10-5 reference, fused	0.804	0.801	0.806
	Trentiles · population, unfused	0.704	0.696	0.713
	Trentiles · population, fused	0.794	0.781	0.806
	Trentiles · 10-10-10 reference, unfused	0.704	0.693	0.712
	Trentiles · 10-10-10 reference, fused	0.789	0.769	0.806
	Centiles · population, unfused	0.698	0.688	0.706
	Centiles · population, fused	0.786	0.772	0.801
Maximum hemoglobin	Deciles · population, unfused	0.702	0.694	0.710
	Deciles · population, fused	0.805	0.800	0.814
	Ventiles · population, unfused	0.705	0.694	0.716
	Ventiles · population, fused	0.797	0.789	0.801
	Ventiles · 5-10-5 reference, unfused	0.702	0.691	0.720
	Ventiles · 5-10-5 reference, fused	0.804	0.801	0.807
	Trentiles · population, unfused	0.699	0.690	0.708
	Trentiles · population, fused	0.795	0.780	0.808
	Trentiles · 10-10-10 reference, unfused	0.698	0.688	0.706
	Trentiles · 10-10-10 reference, fused	0.789	0.768	0.807
	Centiles · population, unfused	0.692	0.683	0.700
	Centiles · population, fused	0.786	0.772	0.801
Vitals (Spearman ρ)				
Minimum heart rate	Deciles · population, unfused	0.445	0.434	0.453
	Deciles · population, fused	0.694	0.690	0.701
	Ventiles · population, unfused	0.439	0.419	0.462
	Ventiles · population, fused	0.694	0.690	0.702
	Ventiles · 5-10-5 reference, unfused	0.443	0.414	0.460
	Ventiles · 5-10-5 reference, fused	0.695	0.693	0.696
	Trentiles · population, unfused	0.441	0.428	0.451
	Trentiles · population, fused	0.695	0.688	0.699

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Trentiles · 10-10-10 reference, unfused	0.439	0.424	0.458
	Trentiles · 10-10-10 reference, fused	0.697	0.694	0.701
	Centiles · population, unfused	0.434	0.419	0.451
	Centiles · population, fused	0.695	0.691	0.698
Maximum heart rate	Deciles · population, unfused	0.451	0.439	0.457
	Deciles · population, fused	0.644	0.640	0.649
	Ventiles · population, unfused	0.452	0.434	0.470
	Ventiles · population, fused	0.645	0.638	0.652
	Ventiles · 5-10-5 reference, unfused	0.457	0.428	0.469
	Ventiles · 5-10-5 reference, fused	0.644	0.639	0.648
	Trentiles · population, unfused	0.453	0.443	0.464
	Trentiles · population, fused	0.645	0.639	0.650
	Trentiles · 10-10-10 reference, unfused	0.451	0.440	0.471
	Trentiles · 10-10-10 reference, fused	0.647	0.643	0.651
	Centiles · population, unfused	0.448	0.441	0.465
	Centiles · population, fused	0.645	0.637	0.651
Minimum systolic blood pressure	Deciles · population, unfused	0.534	0.525	0.546
	Deciles · population, fused	0.583	0.572	0.591
	Ventiles · population, unfused	0.537	0.533	0.541
	Ventiles · population, fused	0.580	0.570	0.590
	Ventiles · 5-10-5 reference, unfused	0.534	0.529	0.541
	Ventiles · 5-10-5 reference, fused	0.582	0.572	0.588
	Trentiles · population, unfused	0.535	0.529	0.544
	Trentiles · population, fused	0.576	0.571	0.581
	Trentiles · 10-10-10 reference, unfused	0.536	0.530	0.541
	Trentiles · 10-10-10 reference, fused	0.574	0.567	0.583
	Centiles · population, unfused	0.533	0.524	0.538
	Centiles · population, fused	0.575	0.564	0.581
Maximum systolic blood pressure	Deciles · population, unfused	0.481	0.469	0.492
	Deciles · population, fused	0.571	0.561	0.582
	Ventiles · population, unfused	0.483	0.474	0.489
	Ventiles · population, fused	0.568	0.549	0.583
	Ventiles · 5-10-5 reference, unfused	0.479	0.466	0.488
	Ventiles · 5-10-5 reference, fused	0.572	0.562	0.580
	Trentiles · population, unfused	0.487	0.482	0.492
	Trentiles · population, fused	0.565	0.549	0.577
	Trentiles · 10-10-10 reference, unfused	0.481	0.474	0.487
	Trentiles · 10-10-10 reference, fused	0.559	0.546	0.570
	Centiles · population, unfused	0.480	0.464	0.492
	Centiles · population, fused	0.557	0.536	0.568

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Table 9 continued from previous page

Outcome	Configuration	Mean	Min	Max
Minimum diastolic blood pressure	Deciles · population, unfused	0.501	0.495	0.507
	Deciles · population, fused	0.557	0.545	0.573
	Ventiles · population, unfused	0.504	0.501	0.506
	Ventiles · population, fused	0.548	0.543	0.554
	Ventiles · 5-10-5 reference, unfused	0.504	0.495	0.509
	Ventiles · 5-10-5 reference, fused	0.549	0.541	0.554
	Trentiles · population, unfused	0.506	0.498	0.516
	Trentiles · population, fused	0.542	0.534	0.554
	Trentiles · 10-10-10 reference, unfused	0.506	0.499	0.512
	Trentiles · 10-10-10 reference, fused	0.546	0.544	0.549
	Centiles · population, unfused	0.502	0.495	0.511
	Centiles · population, fused	0.539	0.531	0.545
Maximum diastolic blood pressure	Deciles · population, unfused	0.377	0.369	0.382
	Deciles · population, fused	0.422	0.413	0.427
	Ventiles · population, unfused	0.376	0.369	0.384
	Ventiles · population, fused	0.418	0.410	0.423
	Ventiles · 5-10-5 reference, unfused	0.379	0.375	0.383
	Ventiles · 5-10-5 reference, fused	0.418	0.409	0.423
	Trentiles · population, unfused	0.377	0.374	0.379
	Trentiles · population, fused	0.414	0.403	0.423
	Trentiles · 10-10-10 reference, unfused	0.378	0.370	0.385
	Trentiles · 10-10-10 reference, fused	0.416	0.407	0.421
	Centiles · population, unfused	0.379	0.371	0.390
	Centiles · population, fused	0.411	0.407	0.421

Table 10. Family-mean AUROC and Spearman ρ for every Experiment 2 configuration. Each cell reports the mean across six trained models (three seeds × two architectures), with min/max in brackets. Member outcomes receive equal weight within each model before the six-model summary.

Configuration	Hospital AUROC	Organ-support AUROC	Laboratory AUROC	Vital-sign AUROC	Electrolyte ρ	Biomarker ρ	Hematology ρ	Vital-sign ρ
Discrete + time tokens	0.890 [0.887, 0.892]	0.959 [0.959, 0.960]	0.785 [0.780, 0.797]	0.701 [0.696, 0.706]	0.471 [0.456, 0.501]	0.454 [0.447, 0.466]	0.731 [0.725, 0.736]	0.473 [0.464, 0.483]
Soft discretization + time tokens	0.889 [0.886, 0.891]	0.959 [0.957, 0.960]	0.783 [0.762, 0.804]	0.701 [0.697, 0.704]	0.460 [0.431, 0.479]	0.447 [0.440, 0.454]	0.723 [0.715, 0.736]	0.469 [0.462, 0.478]
Code-normalized xVal + time tokens	0.839 [0.836, 0.842]	0.902 [0.898, 0.907]	0.729 [0.726, 0.733]	0.647 [0.644, 0.653]	0.346 [0.328, 0.360]	0.377 [0.362, 0.389]	0.568 [0.563, 0.579]	0.379 [0.373, 0.387]
Affine xVal + time tokens	0.838 [0.836, 0.841]	0.902 [0.900, 0.905]	0.734 [0.721, 0.745]	0.648 [0.642, 0.651]	0.360 [0.345, 0.379]	0.378 [0.366, 0.398]	0.574 [0.571, 0.577]	0.377 [0.373, 0.389]
Discrete + event order	0.895 [0.894, 0.897]	0.963 [0.961, 0.965]	0.794 [0.771, 0.808]	0.707 [0.702, 0.712]	0.490 [0.478, 0.500]	0.466 [0.458, 0.472]	0.744 [0.739, 0.748]	0.487 [0.478, 0.500]
Soft discretization + event order	0.895 [0.894, 0.896]	0.963 [0.961, 0.965]	0.797 [0.787, 0.809]	0.707 [0.703, 0.715]	0.493 [0.480, 0.513]	0.460 [0.454, 0.464]	0.741 [0.734, 0.746]	0.482 [0.475, 0.488]
Code-normalized xVal + event order	0.843 [0.841, 0.845]	0.906 [0.904, 0.907]	0.733 [0.726, 0.739]	0.655 [0.650, 0.662]	0.378 [0.361, 0.390]	0.390 [0.378, 0.409]	0.590 [0.576, 0.600]	0.385 [0.377, 0.391]
Affine xVal + event order	0.842 [0.841, 0.845]	0.907 [0.904, 0.908]	0.739 [0.725, 0.752]	0.654 [0.651, 0.657]	0.375 [0.368, 0.380]	0.393 [0.386, 0.399]	0.591 [0.587, 0.592]	0.387 [0.377, 0.397]
Discrete + hospitalization-relative RoPE	0.894 [0.891, 0.896]	0.961 [0.959, 0.963]	0.793 [0.786, 0.804]	0.708 [0.704, 0.714]	0.491 [0.482, 0.500]	0.456 [0.442, 0.470]	0.749 [0.744, 0.753]	0.491 [0.485, 0.497]
Soft discretization + hospitalization-relative RoPE	0.893 [0.892, 0.895]	0.961 [0.960, 0.963]	0.791 [0.776, 0.799]	0.709 [0.704, 0.712]	0.485 [0.469, 0.499]	0.456 [0.451, 0.461]	0.747 [0.740, 0.751]	0.496 [0.486, 0.503]
Code-normalized xVal + hospitalization-relative RoPE	0.841 [0.838, 0.844]	0.906 [0.903, 0.910]	0.741 [0.730, 0.761]	0.653 [0.648, 0.659]	0.376 [0.369, 0.383]	0.391 [0.383, 0.400]	0.598 [0.588, 0.610]	0.386 [0.379, 0.394]
Affine xVal + hospitalization-relative RoPE	0.842 [0.840, 0.843]	0.906 [0.905, 0.910]	0.739 [0.731, 0.746]	0.656 [0.651, 0.658]	0.374 [0.363, 0.383]	0.384 [0.374, 0.390]	0.597 [0.589, 0.603]	0.389 [0.376, 0.405]

Table 11: Outcome-level AUROC and Spearman ρ for every Experiment 2 configuration. Mean, Min, and Max summarize six trained models (three seeds × two architectures).

Outcome	Configuration	Mean	Min	Max
Hospital outcomes (AUROC)				
Hospital mortality in hours 24–48	Discrete + time tokens	0.960	0.953	0.965
	Soft discretization + time tokens	0.960	0.957	0.966
	Code-normalized xVal + time tokens	0.922	0.915	0.926
	Affine xVal + time tokens	0.918	0.913	0.929
	Discrete + event order	0.964	0.960	0.969
	Soft discretization + event order	0.963	0.961	0.966
	Code-normalized xVal + event order	0.922	0.916	0.928
	Affine xVal + event order	0.917	0.911	0.926
	Discrete + hospitalization-relative RoPE	0.963	0.957	0.968
	Soft discretization + hospitalization-relative RoPE	0.961	0.959	0.964
	Code-normalized xVal + hospitalization-relative RoPE	0.920	0.914	0.924
	Affine xVal + hospitalization-relative RoPE	0.920	0.916	0.926
Still hospitalized at hour 48	Discrete + time tokens	0.809	0.806	0.812
	Soft discretization + time tokens	0.808	0.805	0.811
	Code-normalized xVal + time tokens	0.749	0.747	0.751
	Affine xVal + time tokens	0.750	0.749	0.753
	Discrete + event order	0.819	0.818	0.819
	Soft discretization + event order	0.820	0.818	0.820
	Code-normalized xVal + event order	0.757	0.755	0.760
	Affine xVal + event order	0.758	0.758	0.759
	Discrete + hospitalization-relative RoPE	0.816	0.814	0.817
	Soft discretization + hospitalization-relative RoPE	0.814	0.809	0.818
	Code-normalized xVal + hospitalization-relative RoPE	0.754	0.752	0.757
	Affine xVal + hospitalization-relative RoPE	0.755	0.753	0.757
First ICU admission in hours 24–48	Discrete + time tokens	0.833	0.827	0.835
	Soft discretization + time tokens	0.831	0.826	0.834
	Code-normalized xVal + time tokens	0.778	0.771	0.782
	Affine xVal + time tokens	0.776	0.768	0.780
	Discrete + event order	0.839	0.835	0.842
	Soft discretization + event order	0.839	0.836	0.843
	Code-normalized xVal + event order	0.782	0.776	0.789
	Affine xVal + event order	0.784	0.780	0.786

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Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Discrete + hospitalization-relative RoPE	0.840	0.834	0.844
	Soft discretization + hospitalization-relative RoPE	0.839	0.838	0.841
	Code-normalized xVal + hospitalization-relative RoPE	0.782	0.776	0.789
	Affine xVal + hospitalization-relative RoPE	0.782	0.779	0.787
Still in ICU at hour 48	Discrete + time tokens	0.957	0.957	0.958
	Soft discretization + time tokens	0.957	0.955	0.958
	Code-normalized xVal + time tokens	0.909	0.908	0.909
	Affine xVal + time tokens	0.908	0.908	0.909
	Discrete + event order	0.959	0.959	0.959
	Soft discretization + event order	0.959	0.958	0.960
	Code-normalized xVal + event order	0.910	0.909	0.910
	Affine xVal + event order	0.910	0.910	0.911
	Discrete + hospitalization-relative RoPE	0.959	0.958	0.960
	Soft discretization + hospitalization-relative RoPE	0.958	0.958	0.959
	Code-normalized xVal + hospitalization-relative RoPE	0.910	0.909	0.910
	Affine xVal + hospitalization-relative RoPE	0.910	0.909	0.910
Organ support (AUROC)				
Invasive mechanical ventilation	Discrete + time tokens	0.958	0.956	0.960
	Soft discretization + time tokens	0.958	0.956	0.960
	Code-normalized xVal + time tokens	0.902	0.900	0.905
	Affine xVal + time tokens	0.902	0.900	0.906
	Discrete + event order	0.961	0.958	0.963
	Soft discretization + event order	0.962	0.961	0.965
	Code-normalized xVal + event order	0.903	0.901	0.906
	Affine xVal + event order	0.906	0.905	0.907
	Discrete + hospitalization-relative RoPE	0.961	0.960	0.962
	Soft discretization + hospitalization-relative RoPE	0.961	0.958	0.963
	Code-normalized xVal + hospitalization-relative RoPE	0.903	0.901	0.906
	Affine xVal + hospitalization-relative RoPE	0.906	0.902	0.907
Vasopressor initiation	Discrete + time tokens	0.913	0.909	0.917
	Soft discretization + time tokens	0.914	0.912	0.916
	Code-normalized xVal + time tokens	0.841	0.836	0.846
	Affine xVal + time tokens	0.845	0.840	0.848

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Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Discrete + event order	0.917	0.913	0.922
	Soft discretization + event order	0.919	0.913	0.922
	Code-normalized xVal + event order	0.848	0.842	0.852
	Affine xVal + event order	0.849	0.845	0.851
	Discrete + hospitalization-relative RoPE	0.916	0.912	0.921
	Soft discretization + hospitalization-relative RoPE	0.917	0.912	0.921
	Code-normalized xVal + hospitalization-relative RoPE	0.849	0.843	0.856
	Affine xVal + hospitalization-relative RoPE	0.848	0.846	0.850
Continuous renal replacement	Discrete + time tokens	0.981	0.978	0.984
	Soft discretization + time tokens	0.979	0.974	0.982
	Code-normalized xVal + time tokens	0.932	0.920	0.943
	Affine xVal + time tokens	0.930	0.918	0.943
	Discrete + event order	0.988	0.983	0.991
	Soft discretization + event order	0.985	0.981	0.990
	Code-normalized xVal + event order	0.934	0.925	0.940
	Affine xVal + event order	0.937	0.926	0.942
	Discrete + hospitalization-relative RoPE	0.983	0.978	0.988
	Soft discretization + hospitalization-relative RoPE	0.983	0.977	0.990
	Code-normalized xVal + hospitalization-relative RoPE	0.935	0.925	0.941
	Affine xVal + hospitalization-relative RoPE	0.935	0.926	0.944
Hemodialysis initiation	Discrete + time tokens	0.986	0.984	0.987
	Soft discretization + time tokens	0.983	0.977	0.987
	Code-normalized xVal + time tokens	0.933	0.919	0.938
	Affine xVal + time tokens	0.931	0.926	0.935
	Discrete + event order	0.985	0.980	0.989
	Soft discretization + event order	0.984	0.981	0.988
	Code-normalized xVal + event order	0.937	0.932	0.941
	Affine xVal + event order	0.935	0.932	0.937
	Discrete + hospitalization-relative RoPE	0.984	0.981	0.989
	Soft discretization + hospitalization-relative RoPE	0.984	0.980	0.987
	Code-normalized xVal + hospitalization-relative RoPE	0.936	0.930	0.940
	Affine xVal + hospitalization-relative RoPE	0.937	0.930	0.943
Laboratory (AUROC)				

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
Hyperkalemia	Discrete + time tokens	0.719	0.710	0.724
	Soft discretization + time tokens	0.722	0.706	0.737
	Code-normalized xVal + time tokens	0.656	0.638	0.668
	Affine xVal + time tokens	0.667	0.655	0.690
	Discrete + event order	0.724	0.711	0.742
	Soft discretization + event order	0.737	0.712	0.756
	Code-normalized xVal + event order	0.656	0.636	0.684
	Affine xVal + event order	0.672	0.654	0.694
	Discrete + hospitalization-relative RoPE	0.735	0.727	0.743
	Soft discretization + hospitalization-relative RoPE	0.732	0.710	0.745
	Code-normalized xVal + hospitalization-relative RoPE	0.667	0.657	0.684
	Affine xVal + hospitalization-relative RoPE	0.661	0.645	0.672
Severe hypokalemia	Discrete + time tokens	0.709	0.661	0.762
	Soft discretization + time tokens	0.692	0.629	0.751
	Code-normalized xVal + time tokens	0.650	0.632	0.664
	Affine xVal + time tokens	0.654	0.600	0.685
	Discrete + event order	0.734	0.727	0.742
	Soft discretization + event order	0.738	0.712	0.760
	Code-normalized xVal + event order	0.627	0.606	0.661
	Affine xVal + event order	0.665	0.614	0.694
	Discrete + hospitalization-relative RoPE	0.712	0.685	0.735
	Soft discretization + hospitalization-relative RoPE	0.709	0.676	0.745
	Code-normalized xVal + hospitalization-relative RoPE	0.662	0.637	0.718
	Affine xVal + hospitalization-relative RoPE	0.656	0.642	0.681
Severe anemia	Discrete + time tokens	0.861	0.858	0.868
	Soft discretization + time tokens	0.859	0.856	0.865
	Code-normalized xVal + time tokens	0.778	0.769	0.787
	Affine xVal + time tokens	0.784	0.778	0.791
	Discrete + event order	0.865	0.859	0.872
	Soft discretization + event order	0.867	0.863	0.872
	Code-normalized xVal + event order	0.789	0.784	0.793
	Affine xVal + event order	0.789	0.785	0.795
	Discrete + hospitalization-relative RoPE	0.867	0.864	0.874
	Soft discretization + hospitalization-relative RoPE	0.867	0.862	0.871
	Code-normalized xVal + hospitalization-relative RoPE	0.794	0.789	0.799

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Affine xVal + hospitalization-relative RoPE	0.789	0.785	0.794
Hypoglycemia	Discrete + time tokens	0.729	0.719	0.746
	Soft discretization + time tokens	0.730	0.711	0.742
	Code-normalized xVal + time tokens	0.659	0.648	0.672
	Affine xVal + time tokens	0.662	0.641	0.674
	Discrete + event order	0.736	0.724	0.744
	Soft discretization + event order	0.736	0.723	0.745
	Code-normalized xVal + event order	0.675	0.660	0.694
	Affine xVal + event order	0.660	0.652	0.683
	Discrete + hospitalization-relative RoPE	0.732	0.723	0.738
	Soft discretization + hospitalization-relative RoPE	0.739	0.724	0.754
	Code-normalized xVal + hospitalization-relative RoPE	0.670	0.646	0.681
	Affine xVal + hospitalization-relative RoPE	0.672	0.649	0.690
Profound hyponatremia	Discrete + time tokens	0.738	0.704	0.763
	Soft discretization + time tokens	0.739	0.692	0.774
	Code-normalized xVal + time tokens	0.718	0.677	0.739
	Affine xVal + time tokens	0.722	0.668	0.780
	Discrete + event order	0.761	0.681	0.815
	Soft discretization + event order	0.750	0.712	0.783
	Code-normalized xVal + event order	0.733	0.687	0.765
	Affine xVal + event order	0.729	0.664	0.777
	Discrete + hospitalization-relative RoPE	0.759	0.731	0.788
	Soft discretization + hospitalization-relative RoPE	0.745	0.703	0.766
	Code-normalized xVal + hospitalization-relative RoPE	0.738	0.675	0.772
	Affine xVal + hospitalization-relative RoPE	0.737	0.700	0.779
Severe hypernatremia	Discrete + time tokens	0.956	0.903	0.975
	Soft discretization + time tokens	0.958	0.924	0.983
	Code-normalized xVal + time tokens	0.914	0.897	0.924
	Affine xVal + time tokens	0.916	0.895	0.941
	Discrete + event order	0.944	0.907	0.971
	Soft discretization + event order	0.955	0.929	0.974
	Code-normalized xVal + event order	0.921	0.890	0.935
	Affine xVal + event order	0.916	0.895	0.925
	Discrete + hospitalization-relative RoPE	0.953	0.935	0.972
	Soft discretization + hospitalization-relative RoPE	0.952	0.929	0.978

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Code-normalized xVal + hospitalization-relative RoPE	0.918	0.888	0.929
	Affine xVal + hospitalization-relative RoPE	0.916	0.910	0.926
Vitals (AUROC)				
Tachycardia	Discrete + time tokens	0.703	0.690	0.717
	Soft discretization + time tokens	0.700	0.693	0.709
	Code-normalized xVal + time tokens	0.650	0.632	0.663
	Affine xVal + time tokens	0.644	0.626	0.662
	Discrete + event order	0.708	0.696	0.726
	Soft discretization + event order	0.711	0.698	0.722
	Code-normalized xVal + event order	0.661	0.647	0.673
	Affine xVal + event order	0.656	0.644	0.667
	Discrete + hospitalization-relative RoPE	0.709	0.702	0.720
	Soft discretization + hospitalization-relative RoPE	0.713	0.709	0.719
	Code-normalized xVal + hospitalization-relative RoPE	0.656	0.643	0.665
	Affine xVal + hospitalization-relative RoPE	0.659	0.651	0.673
Severe hypertension	Discrete + time tokens	0.656	0.645	0.663
	Soft discretization + time tokens	0.661	0.653	0.670
	Code-normalized xVal + time tokens	0.601	0.595	0.604
	Affine xVal + time tokens	0.609	0.602	0.618
	Discrete + event order	0.665	0.657	0.676
	Soft discretization + event order	0.666	0.658	0.678
	Code-normalized xVal + event order	0.607	0.597	0.620
	Affine xVal + event order	0.609	0.603	0.620
	Discrete + hospitalization-relative RoPE	0.667	0.657	0.676
	Soft discretization + hospitalization-relative RoPE	0.666	0.655	0.676
	Code-normalized xVal + hospitalization-relative RoPE	0.611	0.601	0.620
	Affine xVal + hospitalization-relative RoPE	0.612	0.602	0.620
Hypotension	Discrete + time tokens	0.744	0.738	0.750
	Soft discretization + time tokens	0.743	0.739	0.746
	Code-normalized xVal + time tokens	0.691	0.681	0.697
	Affine xVal + time tokens	0.691	0.683	0.699
	Discrete + event order	0.748	0.742	0.753
	Soft discretization + event order	0.745	0.741	0.748
	Code-normalized xVal + event order	0.696	0.688	0.704
	Affine xVal + event order	0.696	0.691	0.702
	Discrete + hospitalization-relative RoPE	0.748	0.745	0.751

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Soft discretization + hospitalization-relative RoPE	0.749	0.745	0.756
	Code-normalized xVal + hospitalization-relative RoPE	0.693	0.692	0.695
	Affine xVal + hospitalization-relative RoPE	0.696	0.691	0.701
Electrolytes (Spearman ρ)				
Minimum potassium	Discrete + time tokens	0.485	0.469	0.500
	Soft discretization + time tokens	0.482	0.443	0.499
	Code-normalized xVal + time tokens	0.363	0.343	0.378
	Affine xVal + time tokens	0.376	0.359	0.399
	Discrete + event order	0.505	0.497	0.516
	Soft discretization + event order	0.504	0.488	0.512
	Code-normalized xVal + event order	0.387	0.371	0.402
	Affine xVal + event order	0.393	0.384	0.396
	Discrete + hospitalization-relative RoPE	0.506	0.492	0.517
	Soft discretization + hospitalization-relative RoPE	0.495	0.471	0.503
	Code-normalized xVal + hospitalization-relative RoPE	0.394	0.385	0.402
	Affine xVal + hospitalization-relative RoPE	0.394	0.386	0.402
Peak potassium	Discrete + time tokens	0.482	0.465	0.496
	Soft discretization + time tokens	0.479	0.443	0.495
	Code-normalized xVal + time tokens	0.360	0.343	0.374
	Affine xVal + time tokens	0.373	0.357	0.394
	Discrete + event order	0.501	0.492	0.510
	Soft discretization + event order	0.501	0.486	0.509
	Code-normalized xVal + event order	0.383	0.370	0.396
	Affine xVal + event order	0.388	0.380	0.392
	Discrete + hospitalization-relative RoPE	0.503	0.491	0.513
	Soft discretization + hospitalization-relative RoPE	0.493	0.472	0.504
	Code-normalized xVal + hospitalization-relative RoPE	0.390	0.380	0.399
	Affine xVal + hospitalization-relative RoPE	0.388	0.380	0.398
Minimum sodium	Discrete + time tokens	0.462	0.441	0.506
	Soft discretization + time tokens	0.443	0.422	0.474
	Code-normalized xVal + time tokens	0.334	0.317	0.349
	Affine xVal + time tokens	0.348	0.334	0.363
	Discrete + event order	0.481	0.460	0.495
	Soft discretization + event order	0.487	0.458	0.519
	Code-normalized xVal + event order	0.373	0.355	0.387
	Affine xVal + event order	0.361	0.348	0.367

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Discrete + hospitalization-relative RoPE	0.480	0.453	0.494
	Soft discretization + hospitalization-relative RoPE	0.478	0.441	0.498
	Code-normalized xVal + hospitalization-relative RoPE	0.363	0.349	0.377
	Affine xVal + hospitalization-relative RoPE	0.359	0.341	0.379
Maximum sodium	Discrete + time tokens	0.457	0.432	0.502
	Soft discretization + time tokens	0.437	0.417	0.467
	Code-normalized xVal + time tokens	0.328	0.309	0.342
	Affine xVal + time tokens	0.343	0.330	0.359
	Discrete + event order	0.475	0.452	0.491
	Soft discretization + event order	0.482	0.453	0.513
	Code-normalized xVal + event order	0.368	0.350	0.381
	Affine xVal + event order	0.357	0.344	0.365
	Discrete + hospitalization-relative RoPE	0.474	0.445	0.487
	Soft discretization + hospitalization-relative RoPE	0.473	0.433	0.492
	Code-normalized xVal + hospitalization-relative RoPE	0.359	0.346	0.374
	Affine xVal + hospitalization-relative RoPE	0.355	0.337	0.373
Biomarkers (Spearman ρ)				
Minimum creatinine	Discrete + time tokens	0.570	0.560	0.580
	Soft discretization + time tokens	0.565	0.551	0.577
	Code-normalized xVal + time tokens	0.478	0.471	0.489
	Affine xVal + time tokens	0.486	0.470	0.504
	Discrete + event order	0.583	0.574	0.589
	Soft discretization + event order	0.586	0.580	0.594
	Code-normalized xVal + event order	0.488	0.477	0.509
	Affine xVal + event order	0.498	0.493	0.504
	Discrete + hospitalization-relative RoPE	0.583	0.568	0.602
	Soft discretization + hospitalization-relative RoPE	0.583	0.571	0.589
	Code-normalized xVal + hospitalization-relative RoPE	0.492	0.482	0.504
	Affine xVal + hospitalization-relative RoPE	0.496	0.485	0.504
Peak creatinine	Discrete + time tokens	0.573	0.564	0.583
	Soft discretization + time tokens	0.569	0.556	0.580
	Code-normalized xVal + time tokens	0.480	0.473	0.492
	Affine xVal + time tokens	0.489	0.472	0.506
	Discrete + event order	0.586	0.578	0.592
	Soft discretization + event order	0.589	0.583	0.596

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Code-normalized xVal + event order	0.490	0.479	0.510
	Affine xVal + event order	0.500	0.494	0.505
	Discrete + hospitalization-relative RoPE	0.586	0.572	0.604
	Soft discretization + hospitalization-relative RoPE	0.587	0.575	0.592
	Code-normalized xVal + hospitalization-relative RoPE	0.495	0.485	0.506
	Affine xVal + hospitalization-relative RoPE	0.498	0.488	0.505
Minimum troponin	Discrete + time tokens	0.450	0.426	0.466
	Soft discretization + time tokens	0.451	0.439	0.462
	Code-normalized xVal + time tokens	0.392	0.376	0.409
	Affine xVal + time tokens	0.391	0.373	0.420
	Discrete + event order	0.477	0.462	0.502
	Soft discretization + event order	0.467	0.451	0.490
	Code-normalized xVal + event order	0.403	0.389	0.412
	Affine xVal + event order	0.410	0.392	0.428
	Discrete + hospitalization-relative RoPE	0.455	0.441	0.466
	Soft discretization + hospitalization-relative RoPE	0.453	0.443	0.464
	Code-normalized xVal + hospitalization-relative RoPE	0.409	0.399	0.421
	Affine xVal + hospitalization-relative RoPE	0.407	0.395	0.429
Peak troponin	Discrete + time tokens	0.438	0.409	0.461
	Soft discretization + time tokens	0.437	0.422	0.452
	Code-normalized xVal + time tokens	0.382	0.364	0.400
	Affine xVal + time tokens	0.385	0.367	0.406
	Discrete + event order	0.467	0.448	0.488
	Soft discretization + event order	0.453	0.440	0.472
	Code-normalized xVal + event order	0.398	0.383	0.408
	Affine xVal + event order	0.403	0.385	0.419
	Discrete + hospitalization-relative RoPE	0.443	0.421	0.456
	Soft discretization + hospitalization-relative RoPE	0.440	0.421	0.453
	Code-normalized xVal + hospitalization-relative RoPE	0.400	0.391	0.409
	Affine xVal + hospitalization-relative RoPE	0.398	0.381	0.425
Minimum glucose	Discrete + time tokens	0.460	0.450	0.468
	Soft discretization + time tokens	0.458	0.446	0.468
	Code-normalized xVal + time tokens	0.366	0.357	0.376
	Affine xVal + time tokens	0.376	0.368	0.385
	Discrete + event order	0.470	0.464	0.477

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Soft discretization + event order	0.477	0.466	0.483
	Code-normalized xVal + event order	0.383	0.377	0.390
	Affine xVal + event order	0.384	0.379	0.395
	Discrete + hospitalization-relative RoPE	0.468	0.465	0.472
	Soft discretization + hospitalization-relative RoPE	0.469	0.458	0.475
	Code-normalized xVal + hospitalization-relative RoPE	0.385	0.371	0.392
	Affine xVal + hospitalization-relative RoPE	0.382	0.378	0.387
Maximum glucose	Discrete + time tokens	0.503	0.495	0.510
	Soft discretization + time tokens	0.502	0.492	0.510
	Code-normalized xVal + time tokens	0.404	0.396	0.413
	Affine xVal + time tokens	0.413	0.406	0.419
	Discrete + event order	0.514	0.508	0.520
	Soft discretization + event order	0.520	0.511	0.525
	Code-normalized xVal + event order	0.419	0.414	0.426
	Affine xVal + event order	0.421	0.417	0.429
	Discrete + hospitalization-relative RoPE	0.512	0.508	0.515
	Soft discretization + hospitalization-relative RoPE	0.512	0.504	0.517
	Code-normalized xVal + hospitalization-relative RoPE	0.421	0.410	0.428
	Affine xVal + hospitalization-relative RoPE	0.419	0.414	0.423
Minimum natriuretic peptide	Discrete + time tokens	0.322	0.295	0.346
	Soft discretization + time tokens	0.297	0.260	0.317
	Code-normalized xVal + time tokens	0.256	0.208	0.293
	Affine xVal + time tokens	0.245	0.198	0.292
	Discrete + event order	0.317	0.303	0.342
	Soft discretization + event order	0.295	0.264	0.319
	Code-normalized xVal + event order	0.274	0.234	0.316
	Affine xVal + event order	0.271	0.256	0.290
	Discrete + hospitalization-relative RoPE	0.304	0.274	0.349
	Soft discretization + hospitalization-relative RoPE	0.306	0.290	0.319
	Code-normalized xVal + hospitalization-relative RoPE	0.269	0.250	0.290
	Affine xVal + hospitalization-relative RoPE	0.240	0.210	0.260
Peak natriuretic peptide	Discrete + time tokens	0.318	0.295	0.344
	Soft discretization + time tokens	0.293	0.252	0.326

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Code-normalized xVal + time tokens	0.255	0.210	0.291
	Affine xVal + time tokens	0.241	0.200	0.289
	Discrete + event order	0.313	0.304	0.332
	Soft discretization + event order	0.290	0.259	0.315
	Code-normalized xVal + event order	0.263	0.228	0.313
	Affine xVal + event order	0.258	0.242	0.286
	Discrete + hospitalization-relative RoPE	0.298	0.272	0.334
	Soft discretization + hospitalization-relative RoPE	0.300	0.284	0.315
	Code-normalized xVal + hospitalization-relative RoPE	0.259	0.240	0.284
	Affine xVal + hospitalization-relative RoPE	0.231	0.204	0.253
Hematology (Spearman ρ)				
Minimum hemoglobin	Discrete + time tokens	0.733	0.728	0.738
	Soft discretization + time tokens	0.726	0.718	0.738
	Code-normalized xVal + time tokens	0.570	0.565	0.580
	Affine xVal + time tokens	0.576	0.573	0.579
	Discrete + event order	0.746	0.741	0.751
	Soft discretization + event order	0.743	0.736	0.748
	Code-normalized xVal + event order	0.591	0.579	0.601
	Affine xVal + event order	0.592	0.589	0.594
	Discrete + hospitalization-relative RoPE	0.751	0.745	0.755
	Soft discretization + hospitalization-relative RoPE	0.749	0.742	0.753
	Code-normalized xVal + hospitalization-relative RoPE	0.599	0.590	0.611
	Affine xVal + hospitalization-relative RoPE	0.598	0.590	0.604
Maximum hemoglobin	Discrete + time tokens	0.728	0.723	0.734
	Soft discretization + time tokens	0.721	0.712	0.733
	Code-normalized xVal + time tokens	0.566	0.560	0.578
	Affine xVal + time tokens	0.572	0.569	0.574
	Discrete + event order	0.742	0.736	0.746
	Soft discretization + event order	0.739	0.731	0.744
	Code-normalized xVal + event order	0.588	0.574	0.599
	Affine xVal + event order	0.589	0.585	0.591
	Discrete + hospitalization-relative RoPE	0.747	0.742	0.751
	Soft discretization + hospitalization-relative RoPE	0.745	0.738	0.749
	Code-normalized xVal + hospitalization-relative RoPE	0.597	0.586	0.609

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Affine xVal + hospitalization-relative RoPE	0.595	0.587	0.602
Vitals (Spearman ρ)				
Minimum heart rate	Discrete + time tokens	0.452	0.429	0.475
	Soft discretization + time tokens	0.433	0.419	0.459
	Code-normalized xVal + time tokens	0.343	0.332	0.366
	Affine xVal + time tokens	0.345	0.329	0.377
	Discrete + event order	0.463	0.445	0.490
	Soft discretization + event order	0.454	0.434	0.466
	Code-normalized xVal + event order	0.349	0.335	0.360
	Affine xVal + event order	0.349	0.332	0.364
	Discrete + hospitalization-relative RoPE	0.467	0.455	0.490
	Soft discretization + hospitalization-relative RoPE	0.479	0.460	0.495
	Code-normalized xVal + hospitalization-relative RoPE	0.353	0.338	0.379
	Affine xVal + hospitalization-relative RoPE	0.360	0.324	0.393
Maximum heart rate	Discrete + time tokens	0.465	0.451	0.493
	Soft discretization + time tokens	0.452	0.442	0.467
	Code-normalized xVal + time tokens	0.361	0.350	0.378
	Affine xVal + time tokens	0.360	0.350	0.387
	Discrete + event order	0.479	0.463	0.504
	Soft discretization + event order	0.472	0.454	0.481
	Code-normalized xVal + event order	0.367	0.360	0.378
	Affine xVal + event order	0.368	0.349	0.387
	Discrete + hospitalization-relative RoPE	0.483	0.472	0.500
	Soft discretization + hospitalization-relative RoPE	0.493	0.479	0.508
	Code-normalized xVal + hospitalization-relative RoPE	0.368	0.356	0.384
	Affine xVal + hospitalization-relative RoPE	0.378	0.350	0.400
Minimum systolic blood pressure	Discrete + time tokens	0.540	0.533	0.547
	Soft discretization + time tokens	0.538	0.531	0.547
	Code-normalized xVal + time tokens	0.433	0.428	0.443
	Affine xVal + time tokens	0.431	0.423	0.436
	Discrete + event order	0.552	0.548	0.557
	Soft discretization + event order	0.550	0.544	0.555
	Code-normalized xVal + event order	0.441	0.438	0.447
	Affine xVal + event order	0.444	0.434	0.450
	Discrete + hospitalization-relative RoPE	0.555	0.550	0.561

Continued on next page

Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Soft discretization + hospitalization-relative RoPE	0.557	0.551	0.562
	Code-normalized xVal + hospitalization-relative RoPE	0.438	0.430	0.448
	Affine xVal + hospitalization-relative RoPE	0.441	0.433	0.451
Maximum systolic blood pressure	Discrete + time tokens	0.488	0.473	0.500
	Soft discretization + time tokens	0.491	0.483	0.500
	Code-normalized xVal + time tokens	0.376	0.365	0.400
	Affine xVal + time tokens	0.381	0.374	0.388
	Discrete + event order	0.509	0.501	0.518
	Soft discretization + event order	0.503	0.494	0.508
	Code-normalized xVal + event order	0.389	0.373	0.397
	Affine xVal + event order	0.391	0.373	0.401
	Discrete + hospitalization-relative RoPE	0.512	0.506	0.518
	Soft discretization + hospitalization-relative RoPE	0.516	0.508	0.527
	Code-normalized xVal + hospitalization-relative RoPE	0.387	0.377	0.397
	Affine xVal + hospitalization-relative RoPE	0.390	0.367	0.411
Minimum diastolic blood pressure	Discrete + time tokens	0.510	0.505	0.516
	Soft discretization + time tokens	0.509	0.500	0.519
	Code-normalized xVal + time tokens	0.459	0.448	0.469
	Affine xVal + time tokens	0.446	0.443	0.453
	Discrete + event order	0.523	0.514	0.536
	Soft discretization + event order	0.521	0.513	0.526
	Code-normalized xVal + event order	0.458	0.453	0.465
	Affine xVal + event order	0.461	0.447	0.473
	Discrete + hospitalization-relative RoPE	0.530	0.518	0.537
	Soft discretization + hospitalization-relative RoPE	0.528	0.523	0.533
	Code-normalized xVal + hospitalization-relative RoPE	0.462	0.452	0.473
	Affine xVal + hospitalization-relative RoPE	0.461	0.454	0.477
Maximum diastolic blood pressure	Discrete + time tokens	0.383	0.377	0.387
	Soft discretization + time tokens	0.389	0.386	0.392
	Code-normalized xVal + time tokens	0.301	0.295	0.305
	Affine xVal + time tokens	0.298	0.291	0.305
	Discrete + event order	0.394	0.387	0.399
	Soft discretization + event order	0.394	0.380	0.401

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Table 11 continued from previous page

Outcome	Configuration	Mean	Min	Max
	Code-normalized xVal + event order	0.307	0.298	0.313
	Affine xVal + event order	0.308	0.299	0.314
	Discrete + hospitalization-relative RoPE	0.398	0.394	0.406
	Soft discretization + hospitalization-relative RoPE	0.400	0.395	0.405
	Code-normalized xVal + hospitalization-relative RoPE	0.308	0.304	0.311
	Affine xVal + hospitalization-relative RoPE	0.304	0.299	0.312

Table 12. Family-mean AUROC and Spearman ρ for every Experiment 3 configuration. Each cell reports the mean across six trained models (three seeds × two architectures), with min/max in brackets. Member outcomes receive equal weight within each model before the six-model summary.

Configuration	Hospital AUROC	Organ-support AUROC	Laboratory AUROC	Vital-sign AUROC	Electrolyte ρ	Biomarker ρ	Hematology ρ	Vital-sign ρ
Native MIMIC	0.877 [0.875, 0.881]	0.926 [0.923, 0.928]	0.729 [0.706, 0.745]	0.711 [0.706, 0.717]	0.425 [0.415, 0.437]	0.417 [0.405, 0.428]	0.635 [0.627, 0.647]	0.494 [0.488, 0.498]
CLIF 2.1.0	0.886 [0.884, 0.887]	0.917 [0.911, 0.923]	0.735 [0.720, 0.752]	0.743 [0.738, 0.750]	0.390 [0.382, 0.395]	0.423 [0.410, 0.433]	0.663 [0.645, 0.678]	0.593 [0.576, 0.611]

Table 13: Outcome-level AUROC and Spearman ρ for every Experiment 3 configuration. Mean, Min, and Max summarize six trained models (three seeds × two architectures).

Outcome	Configuration	Mean	Min	Max
Hospital outcomes (AUROC)				
Hospital mortality in hours 24–48	Native MIMIC	0.925	0.917	0.932
	CLIF 2.1.0	0.939	0.934	0.942
Still hospitalized at hour 48	Native MIMIC	0.861	0.855	0.866
	CLIF 2.1.0	0.870	0.867	0.873
Still in ICU at hour 48	Native MIMIC	0.844	0.843	0.846
	CLIF 2.1.0	0.850	0.848	0.852
Organ support (AUROC)				
Invasive mechanical ventilation	Native MIMIC	0.904	0.901	0.907
	CLIF 2.1.0	0.914	0.913	0.917
Vasopressor initiation	Native MIMIC	0.871	0.870	0.874
	CLIF 2.1.0	0.877	0.870	0.882
Continuous renal replacement	Native MIMIC	0.964	0.960	0.967

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Table 13 continued from previous page

Outcome	Configuration	Mean	Min	Max
	CLIF 2.1.0	0.962	0.956	0.965
Hemodialysis initiation	Native MIMIC	0.964	0.958	0.968
	CLIF 2.1.0	0.914	0.897	0.929
Laboratory (AUROC)				
Hyperkalemia	Native MIMIC	0.682	0.659	0.707
	CLIF 2.1.0	0.700	0.675	0.725
Severe hypokalemia	Native MIMIC	0.627	0.569	0.668
	CLIF 2.1.0	0.646	0.597	0.680
Severe anemia	Native MIMIC	0.787	0.777	0.799
	CLIF 2.1.0	0.812	0.802	0.839
Hypoglycemia	Native MIMIC	0.716	0.696	0.730
	CLIF 2.1.0	0.676	0.665	0.687
Profound hyponatremia	Native MIMIC	0.675	0.618	0.742
	CLIF 2.1.0	0.690	0.665	0.722
Severe hypernatremia	Native MIMIC	0.886	0.869	0.912
	CLIF 2.1.0	0.886	0.853	0.917
Vitals (AUROC)				
Tachycardia	Native MIMIC	0.713	0.703	0.721
	CLIF 2.1.0	0.757	0.743	0.773
Severe hypertension	Native MIMIC	0.674	0.665	0.682
	CLIF 2.1.0	0.708	0.702	0.716
Hypotension	Native MIMIC	0.747	0.743	0.752
	CLIF 2.1.0	0.764	0.761	0.770
Electrolytes (Spearman ρ)				
Minimum potassium	Native MIMIC	0.434	0.423	0.443
	CLIF 2.1.0	0.359	0.341	0.368
Peak potassium	Native MIMIC	0.469	0.458	0.484
	CLIF 2.1.0	0.431	0.427	0.437
Minimum sodium	Native MIMIC	0.408	0.396	0.426
	CLIF 2.1.0	0.399	0.384	0.410
Maximum sodium	Native MIMIC	0.387	0.377	0.407
	CLIF 2.1.0	0.370	0.352	0.383
Biomarkers (Spearman ρ)				
Minimum creatinine	Native MIMIC	0.563	0.539	0.581
	CLIF 2.1.0	0.550	0.529	0.579
Peak creatinine	Native MIMIC	0.578	0.556	0.594
	CLIF 2.1.0	0.565	0.548	0.592
Minimum troponin	Native MIMIC	0.415	0.399	0.457
	CLIF 2.1.0	0.446	0.430	0.466
Peak troponin	Native MIMIC	0.399	0.376	0.439
	CLIF 2.1.0	0.429	0.419	0.442
Minimum glucose	Native MIMIC	0.437	0.422	0.442
	CLIF 2.1.0	0.424	0.410	0.440
Maximum glucose	Native MIMIC	0.508	0.498	0.514
	CLIF 2.1.0	0.499	0.491	0.512

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Table 13 continued from previous page

Outcome	Configuration	Mean	Min	Max
Minimum natriuretic peptide	Native MIMIC	0.221	0.180	0.287
	CLIF 2.1.0	0.241	0.214	0.286
Peak natriuretic peptide	Native MIMIC	0.217	0.181	0.270
	CLIF 2.1.0	0.226	0.205	0.268
Hematology (Spearman ρ)				
Minimum hemoglobin	Native MIMIC	0.637	0.628	0.650
	CLIF 2.1.0	0.665	0.648	0.679
Maximum hemoglobin	Native MIMIC	0.632	0.625	0.643
	CLIF 2.1.0	0.662	0.642	0.677
Vitals (Spearman ρ)				
Minimum heart rate	Native MIMIC	0.465	0.456	0.477
	CLIF 2.1.0	0.622	0.571	0.673
Maximum heart rate	Native MIMIC	0.491	0.481	0.498
	CLIF 2.1.0	0.611	0.568	0.655
Minimum systolic blood pressure	Native MIMIC	0.574	0.566	0.580
	CLIF 2.1.0	0.631	0.627	0.634
Maximum systolic blood pressure	Native MIMIC	0.516	0.503	0.523
	CLIF 2.1.0	0.598	0.589	0.609
Minimum diastolic blood pressure	Native MIMIC	0.522	0.516	0.527
	CLIF 2.1.0	0.613	0.609	0.618
Maximum diastolic blood pressure	Native MIMIC	0.395	0.387	0.403
	CLIF 2.1.0	0.482	0.474	0.491

Appendix C. Representation Analyses

C.1. Embedding Geometry: Shared and Code-Specific Centile Embeddings

The unfused tokenizer reuses the same 100 centile embeddings for every measurement, whereas fusion creates a separate set for each code. In Figure 2, unfused tokens occupy one shared trajectory and fused measurements form code-specific trajectories. Repeated and rounded measurements produce duplicate quantile boundaries, so the realized number of bins varies by code.

C.2. Linear Decodability of Clinical Boundaries

Population-quantile tokenizers do not use clinical reference intervals to set their bin boundaries. We therefore tested whether next-token training nevertheless ordinally placed the bin embeddings so that a supervised linear probe could recover whether each bin fell within or outside its MIMIC-IV reference interval. For seven laboratory measurements and four granularities, we labeled each bin by its midpoint and fit leave-one-out logistic regression to the corresponding embedding.

Reference-range structure became more recoverable as quantization became finer. No decile or ventile probe exceeded its majority-class baseline, and only the creatinine and blood-gas potassium probes did so

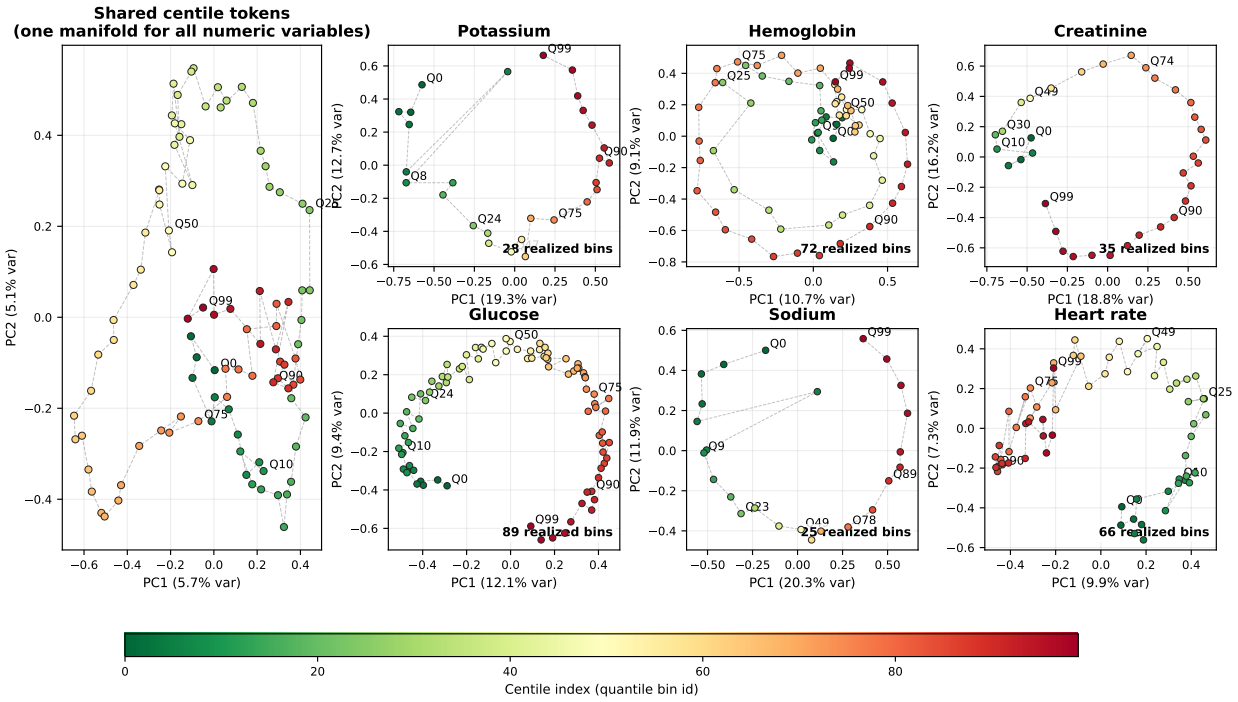

Figure 2. Embedding geometry from Llama training seed 42. Left: principal-component projection of the shared unfused centile embeddings. Right: fused, code-specific centile embeddings for six measurements. Each panel uses the checkpoint selected by validation loss. Panel labels report realized bin counts; potassium contains 28 distinct bins despite a nominal granularity of 100 because rounded values duplicate quantile boundaries.

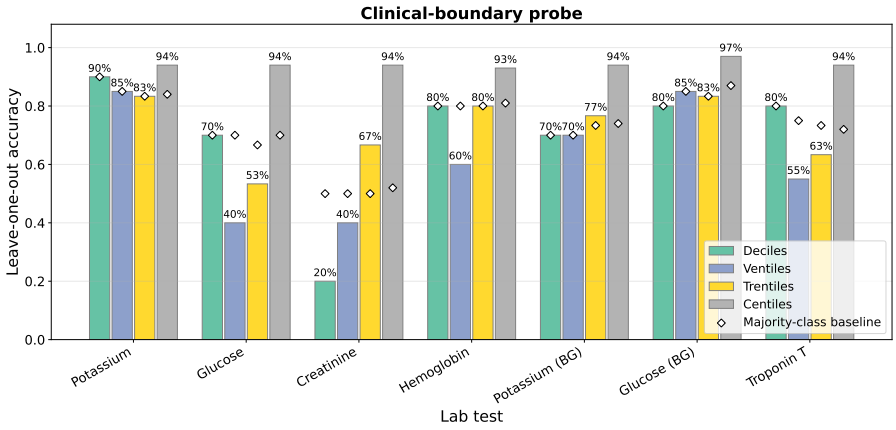

Figure 3. Clinical-boundary probe from Llama training seed 42. Each population-quantile bin is labeled within or outside the MIMIC-IV reference interval according to its midpoint. Bars report leave-one-out logistic-regression accuracy on static bin embeddings from the checkpoint selected by validation loss. Hollow diamonds mark the majority-class baseline: the accuracy obtained by assigning every bin to whichever label, within or outside, is more common for that measurement and granularity. BG denotes blood-gas measurements.

with trentiles. In contrast, all seven centile probes exceeded the baseline by 0.10–0.42 and achieved 93–97% accuracy. Centile embeddings therefore encoded clinically defined normal and abnormal ranges more clearly, although their clearer boundary structure did not yield a consistent downstream advantage in Experiment 1.

Appendix D. Cohort and Data Extraction

We use MIMIC-IV v3.1 (Johnson et al., 2023, 2024) with patient-level train, validation, and test proportions of 70%, 10%, and 20%; all admissions from one patient remain in one partition. Experiments 1–2 include hospital admissions with length of stay of at least 24 hours, yielding 85,057 test admissions. Experiment 3 first restricts to admissions with hospital length of stay of at least 24 hours and at least one linked `icu/icustays` record, then creates a separate patient-level split within that cohort, yielding 16,529 test admissions. The two test sets are therefore not nested. Their different cohort definitions and patient partitions explain why some positive counts are larger in Experiment 3 despite its smaller test set: invasive mechanical ventilation has 1,224 positives in Experiment 3 and 1,153 in Experiments 1–2, while those still in ICU at hour 48 are 8,041 and 8,018, respectively (Table 5). Logistic regression uses a fixed inverse regularization strength of 0.01, and Ridge regression uses a fixed regularization hyperparameter of 100.

Additional Evaluation Metrics. In addition to AUROC, we compute Brier score and expected calibration error using 15 bins for binary outcomes. For regression outcomes, we compute the coefficient of determination (R^2), mean absolute error, and root mean squared error alongside Spearman correlation. Main-text comparisons use family-mean AUROC and Spearman correlation.

Data extraction. Experiments 1–2 use the MEDS extraction pipeline adopted from ETHOS-ARES (Renc et al., 2025), augmented with laboratory reference-range fields. Administrative and billing code tables that may be revised after discharge are excluded to reduce temporal leakage (Ramadan et al., 2025). For tables containing `storetime`, that field determines when information enters the event sequence. Experiment 3 uses the CLIF 2.1.0 conversion of MIMIC-IV (Liao et al., 2026).

Table 14. MIMIC-IV v3.1 patients, hospital admissions, and ICU stays are release totals (Johnson et al., 2024). Laboratory-event rows are counted on the MEDS extraction. Experimental cohorts are subsets.

Statistic	Count
Patients	364,627
Hospital admissions	546,028
ICU stays	94,458
Laboratory events	84,133,368
Unique laboratory codes	895
Codes with reference ranges	334

Appendix E. Training Details and Compute Environment

Additional Training Details. The learning-rate schedule uses linear warmup over the first 1% of optimizer updates. Gradient norms are clipped at 1.0, and no gradient accumulation is used. With unfused population deciles, the Llama 3.2-style decoder has approximately 123.5 million parameters and the Qwen3-style decoder approximately 103.7 million. These counts reflect their respective use of separate and tied input/output embedding matrices and vary with tokenizer vocabulary size. All training runs use one GPU and one Python process. Depending on queue availability, training and hidden-state extraction ran on NVIDIA RTX 6000 Pro or NVIDIA H200 GPUs; tokenization and probes ran on CPU nodes. Checkpoint-and-resume handling was enabled for preemptible jobs. Training jobs request FlashAttention-2 (Dao, 2024) when the `flash_attn` package imports and otherwise use scaled dot-product attention. Kernel selection changes attention efficiency while leaving the training objective unchanged.

Experiment 3 training token exposure for native MIMIC versus CLIF. The full-hospitalization training split contained 684,688,603 native MIMIC tokens and 195,869,100 CLIF tokens before windowing (28.6%). Splitting those sequences into contiguous 4096-context-token windows produced 195,380 native MIMIC and 82,372 CLIF training windows per epoch. Including `TL_CONT` and excluding padding, each epoch provided 684,298,038 native MIMIC and 195,893,782 CLIF tokens. All 12 Experiment 3 runs completed five epochs, processing 4,001,382,400 native MIMIC or 1,686,978,560 CLIF padded input positions per run. Tokenizer vocabulary sizes for native MIMIC and CLIF were 17,752 and 397, respectively.

Appendix F. Tokenized Sequence Length Distributions

Packed-sequence boundaries. Experiment 1 concatenates admissions into fixed-length training blocks. A short, Poisson-distributed run of padding tokens (mean 7) separates adjacent admissions, reducing direct continuity across boundaries while exposing the model to padding during training (Burkhart et al., 2025, 2026).

Appendix G. Formal Definitions of Value Encoders

Soft discretization. Let $b_i < v \leq b_{i+1}$ be the adjacent quantile boundaries surrounding v, and define

$$\alpha = \frac{v - b_i}{b_{i+1} - b_i},$$

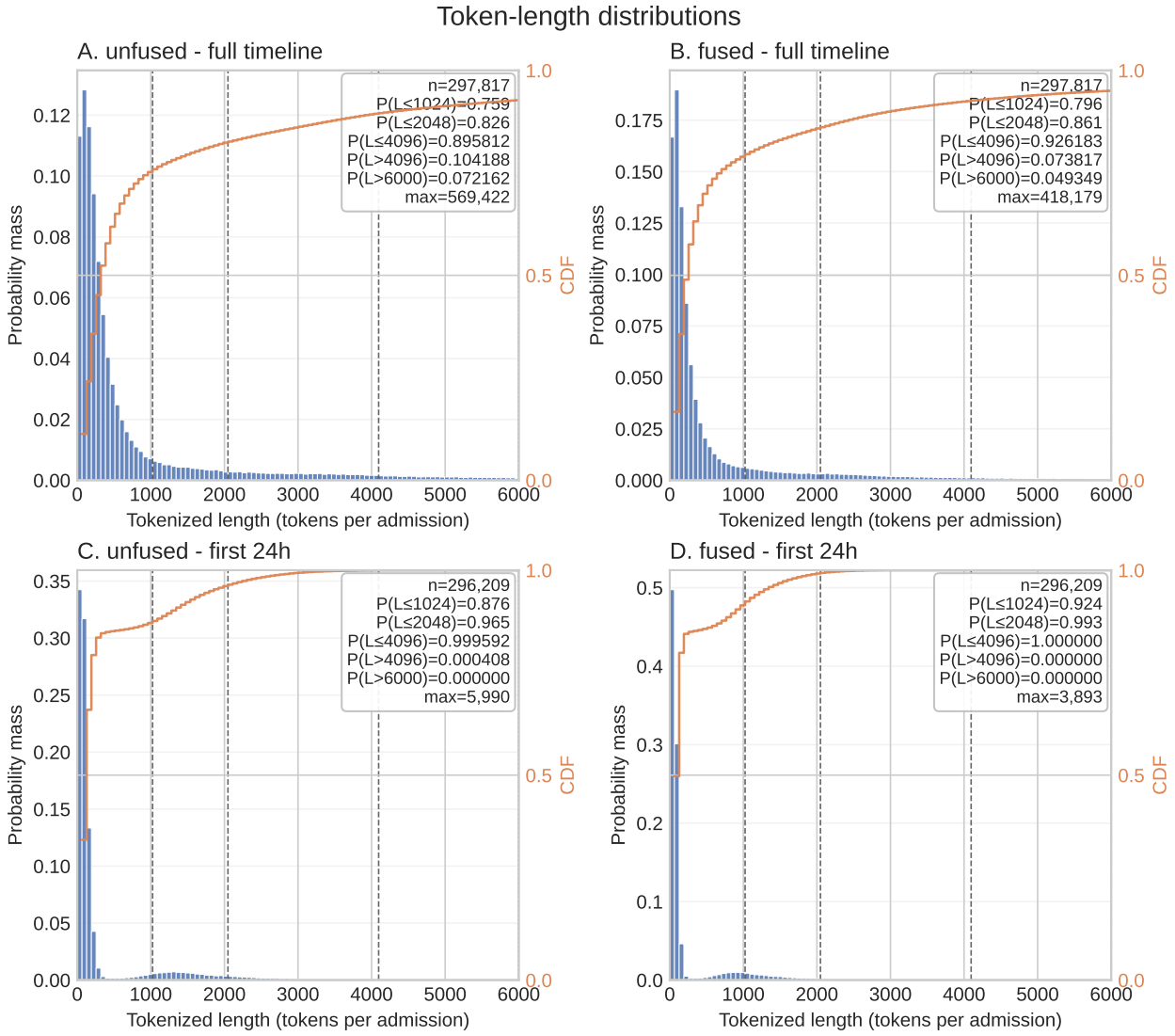

Figure 4. Token lengths for the fused and unfused tokenizers used in Experiments 1–2. Histograms are computed from untruncated training token lists for full and first-24h event streams. Dashed lines mark 1024, 2048, and 4096 tokens. Experiment 3 uses a separate CLIF tokenizer. More than 99.95% of first-24h training admissions have length at most 4096 tokens.

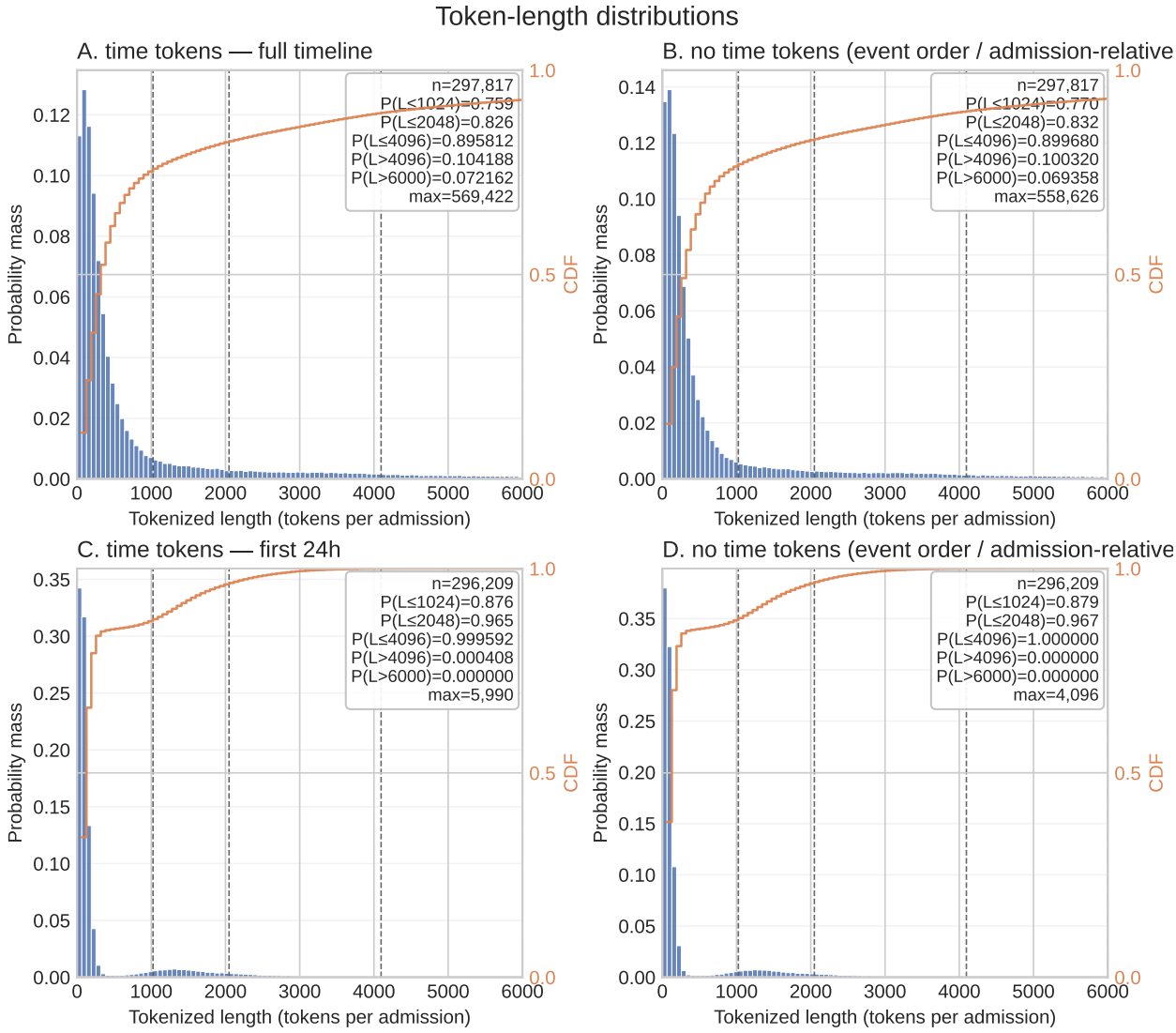

Figure 5. Token lengths with and without explicit time tokens for the tokenizers used in Experiments 1–2. Removing time tokens reduces median first-24h length from 93 to 83 tokens, approximately 11%. Event order and admission-relative RoPE use the same token sequence and differ in the position identifiers supplied during training. Experiment 3 uses a separate CLIF tokenizer.

with $\alpha = 0$ when the boundaries are equal. A value exactly on a stored break b_k remains in bin k, matching the discrete tokenizer's `digitize(..., right=True)`. At a quantile-token target, the same weight defines

$$\mathcal{L}_{\text{soft}}(v) = -(1-\alpha)\log p(Q_i \mid \cdot) - \alpha \log p(Q_{i+1} \mid \cdot).$$

All other targets retain ordinary causal cross-entropy. Values outside the stored boundary range use the first or last embedding.

Code-normalized xVal. For numeric value v from code c, let m_c be the code-specific training median and define the robust scale

$$s_c = \max(\text{IQR}_c / 1.35, 10^{-8}).$$

The normalized value is

$$z = \text{clip}\left(\frac{v - m_c}{s_c}, -5, 5\right).$$

The auxiliary head predicts z at numeric positions. If training summaries are unavailable for a code, numeric scaling and the auxiliary loss are omitted for that event.

Affine xVal. Affine xVal uses the same z and auxiliary head, but represents the value as

$$\mathbf{e}(v) = z\mathbf{e}_{\texttt{[NUM]}} + \mathbf{b},$$

where $\mathbf{b}$ is a learned bias vector. Thus, the code-median value has embedding $\mathbf{b}$ rather than the zero vector.

Appendix H. Numeric Event Families

In Experiments 1–2, the primary MEDS event families that can carry `numeric_value` are laboratory results, vital signs, infusion start rates, subject weight at infusion, and fluid outputs. Their primary sources are `hosp/labevents.valuenum`, `icu/chartevents.valuenum`, `icu/inputevents.rate`, `icu/inputevents.patientweight`, and `icu/outputevents.value`. The extraction also writes `numeric_value` for infusion-end amount and procedure duration when those fields are non-null; the numeric-event rule in Section 3 decides membership. Medication, transfer, ICU admission/discharge, and procedure events remain categorical unless a duration field is present. Reference-range anchoring applies only to laboratory events with valid lower and upper reference fields; other numeric families use population quantiles.

Appendix I. Statistical Comparisons

The main-text tables score eight outcome families on each representation alternative, after averaging outcomes on each of six models (two architectures, three seeds) and then averaging those six scores. A printed family-mean difference is the alternative's family score minus the baseline's. Across three experiments, we vary quantization and fusion, numeric and temporal encoding, native MIMIC versus CLIF 2.1.0. Appendix min/max ranges describe seed and architecture variation. Per-run 95% percentile intervals from 1,000 test-admission bootstraps describe sampling variability for one fitted linear probe. Neither the min/max range nor the per-run interval directly tests a printed family-mean difference.

Each printed difference is tested under the hypothesis that the alternative's and the baseline's predicted scores are exchangeable at the test admission. A paired permutation test uses 1,000 draws and random seed 42. In each draw, the two representations' predicted scores are swapped independently for every test admission with probability one-half; outcome labels are not permuted. The admission-level swap is reused across the six models, across outcomes in the family, and across the eight families in the same table column, so the eight family-level tests in that column become dependent. The family-then-six-model mean is then recomputed on the full test set, without resampling admissions. The two-sided p-value is the fraction of permutations in which the absolute family-mean difference is at least the observed absolute difference.

The baselines for each table are as follows: unfused population deciles for quantization alternatives; unfused tokenizer at the same granularity for each fused arm; discrete bins with inserted time tokens for numeric and temporal alternatives; native MIMIC for CLIF 2.1.0. Then, we apply the Benjamini–Hochberg adjustment at a false discovery rate level of 0.05, respectively within quantization, fusion, numeric encoding, temporal encoding, and schema, not across experiments (Benjamini and Hochberg, 1995). A family-mean difference is called statistically significant when the adjusted p-value is less than 0.05.